Daily And Monthly Revenue Trends

A PROJECT REPORT

Submitted by

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in partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE AND ENGINEERING

Chandigarh University

Department of Computer Science and Engineering

April 2026

BONAFIDE CERTIFICATE

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude and appreciation to all those who have contributed to the successful completion of this project. This project report on "Business Intelligence Dashboard for Sales and Operations Analytics using Microsoft Power BI" has been a rewarding journey, and I am deeply indebted to several individuals for their guidance, support, and encouragement throughout this endeavor. First and foremost, I am extremely grateful to my project supervisor at the Department of Computer Science and Engineering, Chandigarh University, for their invaluable guidance, technical insights, and continuous motivation during the entire course of this project. Their expertise in data analytics and business intelligence proved instrumental in shaping the direction and quality of this work.

I extend my heartfelt thanks to the Head of the Department of Computer Science and Engineering for providing all necessary facilities and resources required to complete this project successfully. The department's commitment to fostering research and practical learning has been a constant source of inspiration.I am also grateful to all the faculty members of the Department of Computer Science and Engineering, Chandigarh University, whose lectures, workshops, and academic input have equipped me with the knowledge and skills required to undertake this project. I would like to acknowledge the contributions of Microsoft Corporation for providing comprehensive documentation and learning resources for Power BI, which proved indispensable throughout the development of this project.My sincere appreciation goes to my fellow students and peers who participated in feedback sessions and provided valuable suggestions for improving the dashboard design and functionality.I am deeply grateful to my family for their unwavering support, patience, and encouragement throughout my academic journey. Their belief in my abilities has been a constant source of strength and motivation.

Finally, I thank the Almighty for bestowing upon me the knowledge, wisdom, and perseverance to complete this project successfully.

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ABSTRACT

In the contemporary era of data-driven decision making, organizations across various industries are increasingly recognizing the critical importance of Business Intelligence (BI) tools for transforming raw data into actionable insights. This project presents the design, development, and implementation of a comprehensive Business Intelligence Dashboard for Sales and Operations Analytics using Microsoft Power BI — one of the most widely adopted BI platforms in the industry today.The primary objective of this project is to develop an interactive and visually compelling Power BI dashboard that enables stakeholders at different organizational levels to monitor key performance indicators (KPIs), track sales trends, analyze operational efficiencies, and derive data-driven insights for strategic decision-making. The dashboard integrates multiple data sources including structured relational databases, Excel spreadsheets, and CSV files, providing a unified and holistic view of organizational performance.

The project encompasses several critical phases: requirement analysis and data gathering, data modeling and transformation using Power Query Editor, development of complex DAX (Data Analysis Expressions) measures and calculated columns, design of interactive visualizations including bar charts, line graphs, pie charts, maps, and custom visuals, and rigorous testing and validation of the implemented solution.Key features of the developed dashboard include real-time sales performance tracking, regional sales distribution analysis, product category performance comparison, customer segmentation and behavior analysis, inventory management insights, and executive-level KPI summary views. The dashboard incorporates advanced features such as row-level security (RLS) for data governance, drill-through functionality for granular analysis, and custom tooltips for enhanced user experience.

The data model follows a star schema design pattern with a central fact table connected to multiple dimension tables, ensuring optimal query performance and accurate calculations. Power Query M language was extensively used for data transformation, cleansing, and preparation, while DAX was employed for complex business logic implementation including time intelligence calculations, ranking functions, and dynamic measures.The results demonstrate that the implemented Power BI dashboard significantly enhances data visibility, reduces reporting time by approximately 70%, and provides decision-makers with intuitive access to

critical business metrics. The interactive nature of the dashboard enables users to slice and filter data across multiple dimensions, facilitating ad-hoc analysis without requiring technical expertise.

Keywords: Business Intelligence, Power BI, Data Analytics, Dashboard, DAX, Power Query, KPI, Data Visualization, Sales Analytics, Data Modeling.

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CHAPTER 1: INTRODUCTION

The digital transformation wave sweeping across global industries has fundamentally reshaped the way organizations capture, process, and utilize data to gain a competitive advantage. In today's data-driven environment, businesses generate vast amounts of information from multiple sources such as transactions, customer interactions, and operational systems. However, the true value of this data lies not in its volume but in the organization's ability to transform it into meaningful insights. In this era of information abundance, organizations that can quickly analyze and interpret data are better positioned to make informed decisions, respond to market changes, and outperform their competitors. As a result, the ability to convert raw data into actionable intelligence has become a key differentiator between market leaders and laggards. To address this growing need, Business Intelligence (BI) tools have emerged as essential components of modern organizational strategies. These tools enable users to collect, integrate, analyze, and visualize data in a structured and efficient manner. BI platforms empower stakeholders at all levels—from executives to operational staff—to access relevant information and make data-driven decisions. By providing interactive dashboards, real-time reporting, and advanced analytics capabilities, BI tools enhance transparency, improve efficiency, and support strategic planning. They also promote a culture of data-driven decision-making, where insights are derived from reliable data rather than assumptions or intuition.

Among the various BI tools available in the market, Microsoft Power BI has gained significant prominence as a leading data visualization and analytics platform. Since its launch in 2015, Power BI has continuously evolved, incorporating advanced features and maintaining a strong position as a Leader in Gartner's Magic Quadrant for Analytics and Business Intelligence Platforms. Its popularity can be attributed to its powerful data connectivity options, which allow users to integrate data from a wide range of sources, including databases, cloud services, and spreadsheets. Additionally, its intuitive drag-and-drop interface makes it accessible to both technical and non-technical users, reducing the learning curve and encouraging widespread adoption. Another important strength of Power BI is its robust Data Analysis Expressions (DAX) language, which enables users to create complex calculations and derive meaningful metrics. Combined with seamless integration within the Microsoft ecosystem—such as Excel, Azure, and SharePoint—Power BI provides a comprehensive solution for organizations seeking to modernize their analytics capabilities. This combination of flexibility, scalability, and

ease of use makes it an ideal choice for organizations aiming to democratize data access and foster a data-driven culture. This project focuses on the design and development of a comprehensive Business Intelligence dashboard using Microsoft Power BI. The primary objective is to provide the organization with a unified analytics platform that supports monitoring and analysis of key business areas, including sales performance, operational efficiency, customer behavior, and financial health. By consolidating data from multiple sources into a single platform, the dashboard eliminates data silos and ensures consistency in reporting.

Furthermore, the dashboard leverages interactive visualizations, filters, and drill-down capabilities to enable users to explore data dynamically and uncover insights quickly. It serves as a single source of truth, ensuring that all stakeholders rely on the same accurate and up-to-date information. Ultimately, this BI solution enhances decision-making by providing timely, relevant, and actionable insights, thereby contributing to improved organizational performance and long-term business success.

1.1 Identification of Client / Need / Relevant Contemporary Issue

The contemporary business landscape is characterized by exponential data growth, with organizations generating and collecting data at an unprecedented rate. According to IDC's Global DataSphere forecast, the amount of data created, captured, copied, and consumed globally is projected to reach 175 zettabytes by 2025, compared to 33 zettabytes in 2018. This data explosion presents both immense opportunities and significant challenges for organizations seeking to extract value from their data assets.

Despite this data abundance, many organizations struggle with what is often termed the 'data-to-insight gap' — the disconnect between the availability of data and the ability to derive actionable insights from it in a timely manner. Traditional reporting approaches, typically built around static Excel spreadsheets and scheduled batch reports, are increasingly inadequate for meeting the dynamic information needs of modern businesses. Key issues with these traditional approaches include:

- Data silos: Information scattered across multiple disconnected systems without a unified view
- Reporting latency: Reports are generated on fixed schedules rather than reflecting real-time data

- Limited interactivity: Static reports cannot be filtered or drilled-down by end users without technical intervention
- High IT dependency: Every new report or modification requires involvement of IT or data engineering teams
- Inconsistent metrics: Different departments using different definitions for the same KPIs
- Scalability limitations: Traditional tools struggle to handle large datasets efficiently

The global Business Intelligence market reflects this need, with a market size valued at USD 29.42 billion in 2023 and projected to grow at a compound annual growth rate (CAGR) of 9.0% from 2024 to 2030. The client for this project represents a mid-sized organization operating in a competitive market where the ability to rapidly respond to market trends, customer preferences, and operational challenges directly impacts business performance.

1.2 Identification of Problem

The core problem addressed by this project is the absence of an integrated, interactive, and real-time Business Intelligence solution that consolidates data from multiple organizational data sources and presents actionable insights through intuitive visualizations accessible to stakeholders across the organization.

1.2.1 Data Fragmentation Problem

Organizational data resides in multiple disparate sources including transactional databases, CRM systems, ERP platforms, Excel files, and external data feeds. The absence of data integration results in inconsistent, incomplete, and often contradictory views of business performance.

1.2.2 Analytical Capability Gap

Business users within the organization lack the technical skills and tools necessary to independently explore and analyze data beyond pre-defined reports. The dependency on IT for every analytical request creates significant bottlenecks, slowing down the decision-making process and reducing organizational agility.

1.2.3 Performance Monitoring Deficiency

The organization lacks a systematic mechanism for continuously monitoring KPIs against targets and benchmarks. Without automated performance alerts and dashboards, deteriorating trends often go unnoticed until they have already caused significant business impact.

1.2.4 Reporting Inefficiency

The current manual reporting process within the organization is not only time-consuming but also highly inefficient and prone to human error. Preparing reports manually requires analysts and managers to collect data from multiple sources, clean and organize it, and then compile it into meaningful formats. This repetitive process consumes a significant amount of time on a daily and monthly basis, reducing overall productivity. Moreover, manual handling of data increases the chances of errors such as incorrect calculations, data duplication, or omission of critical information, which can negatively impact the accuracy of reports.In addition to these challenges, the manual reporting system limits the ability of employees to focus on higher-value analytical tasks. Instead of spending time deriving insights, identifying trends, or supporting strategic decisions, valuable human resources are occupied with routine data preparation activities. This not only slows down the decision-making process but also affects the organization's ability to respond quickly to changing business conditions. Therefore, there is a strong need to automate the reporting process through a centralized Business Intelligence solution. By reducing manual effort and minimizing errors, the organization can improve efficiency, enhance data accuracy, and enable employees to focus on more meaningful and value-driven analytical work.

1.3 Identification of Tasks

Table 1.1: Task Identification Matrix

Task No.	Task Description	Phase	Duration
T1	Requirement gathering and stakeholder interviews	Planning	1 week
T2	Data source identification and access configuration	Planning	1 week
T3	Literature review and technology evaluation	Research	2 weeks
T4	Data model design (star schema)	Design	1 week

Task No.	Task Description	Phase	Duration
T5	Power Query data transformation and cleaning	Development	2 weeks
T6	DAX measures and calculated columns development	Development	2 weeks
T7	Dashboard layout and visualization design	Development	2 weeks
T8	Row-level security implementation	Development	1 week
T9	User acceptance testing	Testing	1 week
T10	Performance optimization	Testing	1 week
T11	Documentation and report preparation	Closure	2 weeks

1.4 Timeline

Table 1.2: Project Timeline with Phases and Deliverables

Week	Phase	Key Activities	Deliverable
1-2	Initiation & Planning	Stakeholder meetings, requirement gathering, scope definition	Project Charter, Requirements Document
3-4	Research & Analysis	Literature review, technology evaluation, BI tool comparison	Literature Review Report
5-6	Data Preparation	Data source connectivity, Power Query transformations, data cleaning	Cleaned Data Model
7-8	Data Modeling	Star schema design, relationships, DAX measures	Data Model with Measures
9-11	Dashboard Development	Visualization design, page layouts, interactivity features	Draft Dashboard
12	Security & Governance	RLS implementation, access control, data validation	Secured Dashboard
13-14	Testing & Refinement	UAT, performance testing, bug fixes, optimization	Tested Dashboard
15-16	Documentation & Closure	Report writing, presentation preparation, project review	Final Report & Presentation

1.5 Organization of the Report

This report is organized into five chapters, each covering a distinct aspect of the project:

- Chapter 1 - Introduction: Provides the background, problem context, client need, identification of tasks, project timeline, and overall organization of the report.

- Chapter 2 - Literature Review / Background Study: Presents a comprehensive review of existing literature on Business Intelligence, data analytics, and visualization techniques.
- Chapter 3 - Design Flow / Process: Details the design methodology employed in the project, including requirement specification, design constraints, alternative designs considered, and the implementation methodology.
- Chapter 4 - Results Analysis and Validation: Presents the implementation results, including data model details, DAX measures developed, dashboard screenshots and descriptions, testing results, and performance analysis.
- Chapter 5 - Conclusion and Future Work: Summarizes the key findings, achievements, and contributions of the project and outlines directions for future enhancement.

CHAPTER 2: LITERATURE REVIEW / BACKGROUND STUDY

This chapter provides a comprehensive review of the existing body of knowledge related to Business Intelligence (BI), data analytics dashboards, and visualization tools, forming the theoretical foundation for the present project. The objective of this review is to understand the evolution, current trends, and technological advancements in the field of BI, as well as to identify best practices and gaps that can be addressed through the proposed solution[1]. By analyzing both academic literature and industry practices, this chapter establishes a clear context for the development of an effective and modern BI dashboard. The review begins with an exploration of the historical evolution of Business Intelligence. Initially, BI systems were primarily focused on static reporting and basic data summarization[2]. Early tools provided limited interactivity and were often dependent on manual data extraction and processing. Over time, advancements in data storage technologies, such as data warehousing and cloud computing, enabled organizations to handle larger volumes of data more efficiently. This evolution led to the emergence of modern BI platforms that support real-time analytics, interactive dashboards, and self-service capabilities[3]. These developments have significantly transformed BI from a reporting tool into a strategic asset for organizations.

In addition to historical analysis, the chapter examines existing BI solutions and their functionalities. Various tools such as Microsoft Power BI, Tableau, and Qlik Sense are evaluated based on their features, usability, scalability, and integration capabilities[4]. These tools have played a crucial role in enabling organizations to visualize complex datasets and derive meaningful insights. The review highlights how modern BI platforms offer advanced features such as drag-and-drop interfaces, real-time data updates, and customizable dashboards, which enhance user experience and analytical efficiency[5]. However, challenges such as data integration complexities, performance issues with large datasets, and the need for proper data governance are also identified. A comparative evaluation of different BI tools is also included to understand their strengths and limitations[6]. Factors such as cost, ease of use, data connectivity, visualization capabilities, and support for advanced analytics are considered in this comparison[7]. This evaluation helps in justifying the selection of Microsoft Power BI as the preferred tool for this project, given its strong integration with the Microsoft ecosystem, robust analytical capabilities, and user-friendly interface[8].

Furthermore, the chapter incorporates a bibliometric analysis of key research contributions in the field of Business Intelligence and data visualization. By analyzing published research papers, journals, and conference proceedings, the review identifies major themes, trends, and emerging areas of interest[9]. This includes topics such as big data analytics, predictive modeling, data storytelling, and the role of artificial intelligence in BI systems. The bibliometric approach provides insights into how the field has evolved over time and highlights the increasing importance of data-driven decision-making in modern organizations[10]. In conclusion, this chapter synthesizes the knowledge gained from literature and existing solutions to provide a solid foundation for the project[11]. It not only highlights the significance of Business Intelligence in today's data-centric environment but also identifies the need for efficient, scalable, and user-friendly BI solutions[12].The insights derived from this review guide the design and development of the proposed Power BI dashboard, ensuring that it aligns with current best practices and technological advancements in the field[13].

2.1 Timeline of the Reported Problem

2.1.1 Early Data Processing Era (1960s-1970s)

The foundations of what we today call Business Intelligence were laid in the 1960s with the advent of Management Information Systems (MIS). These early systems were primarily transaction processing systems that produced standardized operational reports. The concept of Decision Support Systems (DSS) emerged in the 1970s, pioneered by researchers such as Peter Keen and Michael Scott Morton.

2.1.2 Data Warehousing Revolution (1980s-1990s)

The 1980s and 1990s witnessed the emergence of data warehousing as a critical infrastructure component for organizational analytics. Bill Inmon, often referred to as the 'Father of Data Warehousing,' introduced the concept of the data warehouse in 1990. Ralph Kimball's dimensional modeling methodology, introduced in the mid-1990s, provided practical guidance for designing data warehouses using star schemas and snowflake schemas — concepts that remain foundational to modern BI implementations.

2.1.3 BI Democratization (2000s-2010s)

The 2000s marked a significant shift in the BI landscape with the emergence of self-service BI tools. The introduction of Tableau in 2003 marked a pivotal moment, demonstrating that powerful visual analytics could be delivered through an intuitive drag-and-drop interface accessible to non-technical users.

2.1.4 Cloud BI and Modern Analytics Era (2015-Present)

The mid-2010s saw the emergence of cloud-based BI platforms that offered scalability, reduced infrastructure costs, and enhanced collaboration capabilities. Microsoft Power BI, launched in 2015, quickly gained traction due to its deep integration with the Microsoft ecosystem. Today's BI landscape is characterized by the convergence of traditional analytics with artificial intelligence and machine learning.

2.2 Existing Solutions

2.2.1 Microsoft Power BI

Microsoft Power BI is a cloud-based Business Intelligence service that provides non-technical users with tools for aggregating, analyzing, visualizing, and sharing data. Key strengths include seamless integration with Microsoft's ecosystem, an extensive library of over 100 native data connectors, the powerful DAX formula language, and a competitive pricing model.

2.2.2 Tableau

Tableau, now part of Salesforce, is recognized as a pioneer in the data visualization space and remains one of the most powerful tools for creating sophisticated visualizations. Tableau's VizQL technology enables users to create visualizations without writing code. While powerful, it typically comes at a significantly higher price point than Power BI.

2.2.3 Qlik Sense

Qlik Sense is distinguished by its associative data model, which differs fundamentally from the query-based approach used by most BI tools. Qlik's in-memory associative engine enables users to freely explore data relationships without predefined drill paths.

2.2.4 Looker (Google Cloud)

Looker, acquired by Google in 2020, takes a distinctive approach to BI through its LookML modeling layer, which enables data teams to define metrics and data relationships in a reusable, version-controlled manner.

2.2.5 Comparative Analysis of BI Tools

Table 2.2: Comparison of Leading BI Tools — Feature Matrix

Feature	Power BI	Tableau	Qlik Sense	Looker
Ease of Use	High	Medium-High	Medium	Medium
Visualization Quality	High	Very High	High	Medium
Data Modeling	Very High	Medium	High	Very High
Pricing (Entry)	Free/~$10/mo	~$70/mo	~$30/mo	Custom
Cloud Native	Yes	Partial	Partial	Yes
AI/ML Integration	High	Medium	Medium	Medium
Microsoft Integration	Excellent	Good	Good	Fair
Mobile Support	Excellent	Good	Good	Fair
Community Size	Very Large	Large	Medium	Medium
Real-time Data	Good	Good	Excellent	Good

2.3 Bibliometric Analysis

A systematic bibliometric analysis was conducted to evaluate the volume and quality of academic research related to Business Intelligence, Power BI, data dashboards, and analytics-driven decision making. The analysis covered publications from major academic databases including IEEE Xplore, ACM Digital Library, Springer, Elsevier, and Google Scholar, spanning the period from 2015 to 2024.

Table 2.1: Bibliometric Analysis Summary

Author(s)	Year	Title/Topic	Key Contribution
Wixom & Watson	2010	The BI-Based Organization	Framework for BI organizational impact
Eckerson	2010	Performance Dashboards	Dashboard design best practices
Foshay & Kuziemsky	2014	BI Implementation Framework	BI implementation factors
Sacu & Spruit	2015	BIDM Model for BI Development	BI development methodology
Pappas et al.	2018	Big Data and BI Strategy	Big data integration with BI
Santos & Ramos	2019	Power BI for Academic BI	Power BI in educational BI
Larson & Chang	2016	Overview of Self-Service BI	Self-service BI review
Trieu	2017	How Value from BI Systems	Value creation through BI
Chen et al.	2021	DAX Query Performance	Optimization techniques for DAX
Williams & Williams	2021	Profit Impact of BI	ROI analysis of BI investments

2.4 Review Summary

The comprehensive literature review conducted for this project has yielded several critical insights that directly informed the design and development approach:

- Dashboard design principles emphasize cognitive efficiency — minimizing the mental effort required for users to extract insights from visualizations.
- Star schema data modeling remains the dominant approach for BI data models due to its simplicity, query performance advantages, and ease of understanding.
- Self-service BI implementations are most successful when accompanied by strong data governance frameworks that ensure metric consistency and data quality.
- Power BI's DAX language, while powerful, has a significant learning curve. Investing early in a well-designed measure framework pays dividends in efficiency and maintainability.
- User adoption is a critical success factor. Implementations involving end users early and providing adequate training consistently demonstrate higher adoption rates.

2.5 Problem Definition

Based on the needs assessment and literature review conducted during the initial phase of this project, the core problem addressed can be formally defined as the absence of a centralized and interactive Business Intelligence (BI) platform within the organization. Currently, the organization relies on multiple disparate data sources and manual reporting processes, which leads to inefficiencies, inconsistencies, and delays in decision-making. The lack of integration among these data sources prevents stakeholders from obtaining a unified and accurate view of business performance, thereby limiting the organization’s ability to make timely and informed decisions. One of the primary issues identified is the fragmentation of data across various systems such as sales, finance, and operations. These systems often operate independently, resulting in data silos that hinder effective analysis. As a result, users are required to manually extract and combine data from different sources, which is not only time-consuming but also prone to errors. This manual approach increases the risk of inconsistencies, as different teams may apply varying methods and assumptions when preparing reports. Consequently, there is no single source of truth for key business metrics, leading to discrepancies in reported figures and reduced trust in the data. Another significant problem is the absence of standardized business

logic for calculating key performance indicators (KPIs). In the current environment, different departments may define and compute metrics such as revenue, profit, and growth in different ways. This lack of standardization creates confusion and makes it difficult to compare performance across departments or time periods. Without consistent definitions and calculations, it becomes challenging for decision-makers to rely on the data for strategic planning and performance evaluation.

Furthermore, the organization lacks an interactive and user-friendly reporting interface that can present insights in a clear and intuitive manner. Existing reports are often static, complex, and difficult to interpret, especially for non-technical users. This limits the accessibility of data insights to a small group of skilled analysts, while other stakeholders are unable to fully leverage the available information. The absence of interactive visualizations such as dashboards, filters, and drill-down capabilities further restricts users from exploring the data and uncovering meaningful patterns. Additionally, the current reporting system does not support real-time or near real-time data updates, which results in outdated information being used for decision-making. In a dynamic business environment, delays in data availability can lead to missed opportunities and ineffective responses to emerging trends or issues.

Therefore, this project aims to address these challenges by developing a centralized Business Intelligence platform using Power BI. The proposed solution integrates data from multiple sources, applies standardized business logic through consistent calculations, and delivers actionable insights through intuitive and interactive visualizations. This platform is designed to be accessible to stakeholders at all levels of the organization, enabling them to make data-driven decisions efficiently and effectively.

2.6 Goals / Objectives

The project is guided by the following specific, measurable, achievable, relevant, and time-bound objectives:

- Develop a unified data model integrating sales transactions, customer data, product catalog, and regional data into a coherent star schema by end of Week 8.
- Implement a minimum of 30 DAX measures covering sales KPIs, time intelligence calculations, ranking functions, and trend analysis by end of Week 10.

- Create at least 5 interactive dashboard pages with a minimum of 8 visualizations per page by end of Week 11.
- Implement row-level security covering at minimum 3 security roles by end of Week 12.
- Achieve a dashboard page load time of under 10 seconds and pass all defined test cases with a success rate of at least 95% by end of Week 14.
- Deliver complete project documentation including technical specifications, user guide, and this project report by end of Week 16.

CHAPTER 3: DESIGN FLOW / PROCESS

This chapter details the design methodology and process followed in developing the Power BI Business Intelligence Dashboard. It covers the evaluation and selection of specifications and features, design constraints, comparative analysis of design alternatives, the selected design approach, and the detailed implementation methodology.

3.1 Evaluation & Selection of Specifications / Features

3.1.1 Functional Requirements

Table 3.1: Functional Requirements Specification

Req. ID	Requirement	Priority	Source
FR-01	Display total sales, profit, and quantity KPIs on executive summary	Must Have	Management
FR-02	Sales trend visualization by day/week/month/quarter/year	Must Have	Sales Team
FR-03	Regional sales performance map visualization	Must Have	Sales Management
FR-04	Product category and sub-category performance analysis	Must Have	Product Team
FR-05	Customer segmentation and top customer analysis	Should Have	Marketing
FR-06	Sales target vs. actual comparison with variance	Must Have	Management
FR-07	Year-over-year and month-over-month growth calculations	Must Have	Finance
FR-08	Drill-through from summary to transaction detail	Should Have	Analysts
FR-09	Cross-filtering between all visualizations on a page	Must Have	All Users
FR-10	Export to PDF and PowerPoint	Should Have	Management
FR-11	Row-level security by region/territory	Must Have	IT Security
FR-12	Mobile-responsive layout	Could Have	Field Sales
FR-13	Custom tooltips with additional context	Should Have	Analysts
FR-14	Bookmarks for saved views/states	Could Have	Power Users
FR-15	Automated data refresh schedule	Must Have	All Users

3.1.2 Non-Functional Requirements

Table 3.2: Non-Functional Requirements

Req. ID	Requirement	Specification
NFR-01	Performance	Dashboard page load time < 10 seconds
NFR-02	Usability	Navigable by non-technical users without training
NFR-03	Scalability	Support up to 10 million rows in fact table
NFR-04	Availability	99.5% uptime during business hours
NFR-05	Security	Role-based data access with RLS
NFR-06	Data Freshness	Data refreshed minimum twice daily
NFR-07	Compatibility	Compatible with Power BI Desktop and Service
NFR-08	Maintainability	Well-documented measures and transformations

3.2 Design Constraints

3.2.1 Technical Constraints

The design and development of the Power BI dashboard was subject to several technical, organizational, and regulatory constraints. Power BI Service pricing tier limits the use of certain premium features such as paginated reports, incremental refresh, and deployment pipelines. Data volume constraints impose a 1GB per dataset limit for Power BI Pro imports, requiring efficient data modeling. DirectQuery mode, while enabling real-time data, imposes limitations on DAX measure complexity and visualization types.

3.2.2 Organizational Constraints

The project timeline of 16 weeks limits the scope of features implementable in the initial release. The development team consists of a single developer, requiring feature prioritization and scope management. Stakeholder availability for feedback and testing is limited, requiring efficient review cycles.

3.2.3 Standards and Compliance

Data handling must comply with applicable data protection regulations. Report design should adhere to organizational brand guidelines. Dashboard content must not expose personally identifiable information (PII) beyond role-authorized views.

Table 3.3: Design Constraints Analysis

Constraint	Category	Impact	Mitigation
1GB dataset size limit	Technical	Limits data volume	Aggressive aggregation in Power Query
Import vs DirectQuery	Technical	Performance vs real-time tradeoff	Import mode with scheduled refresh
16-week timeline	Organizational	Scope limitation	MVP approach with phased delivery
Data access control	Security	RLS implementation required	Role hierarchy design
<10 sec load time	Performance	Measure optimization needed	Query folding, aggregation tables

3.3 Analysis of Features and Finalization Subject to Constraints

Table 3.4: Feature Prioritization Matrix (MoSCoW)

Feature	Initial Priority	Final Priority
Core KPI Cards (Sales, Profit, Qty)	Must Have	Must Have - Included
Time Intelligence Measures (YoY, MoM)	Must Have	Must Have - Included
Regional Map Visualization	Must Have	Must Have - Included
Product Performance Analysis	Must Have	Must Have - Included
Customer Analysis	Should Have	Should Have - Anonymized
Row-Level Security	Must Have	Must Have - Included
Drill-Through Pages	Should Have	Should Have - Included
Mobile Layout	Could Have	Won't Have - Phase 2
AI Visuals (Decomposition Tree)	Should Have	Could Have - Basic only
Paginated Reports	Could Have	Won't Have - Phase 2
Real-time Streaming	Could Have	Won't Have - Phase 2

3.4 Design Flow

3.4.1 Design Alternative 1: Flat Model Approach

This approach involves creating a denormalized, flat table model where all data attributes are combined into a single wide table. Advantages include simpler model structure, no relationship complexity, and ease of understanding for beginners. Disadvantages include significant data redundancy, poor performance with large datasets, increased DAX measure complexity, and difficulty in maintenance and extension.

3.4.2 Design Alternative 2: Star Schema Model

This approach follows Kimball's dimensional modeling methodology, creating a central fact table containing transactional data surrounded by dimension tables containing descriptive attributes. Advantages include optimal query performance, minimal data redundancy, clear separation of facts and dimensions, easier DAX development, and high extensibility.

Table 3.5: Design Alternative Comparison

Criterion	Flat Model	Star Schema	Winner
Query Performance	Poor at scale	Excellent	Star Schema
Storage Efficiency	Low (redundancy)	High	Star Schema
DAX Complexity	High	Low-Medium	Star Schema
Extensibility	Poor	Excellent	Star Schema
Initial Setup Complexity	Low	Medium	Flat Model
Maintenance	Difficult	Easy	Star Schema
Industry Best Practice	Not recommended	Recommended	Star Schema

3.5 Design Selection

Based on the comparative analysis, the Star Schema Data Model approach was selected as the optimal design for this project. The selected star schema design comprises the following tables:

Fact Tables:

- **Fact_Sales:** Central fact table containing sales transaction records with foreign keys to all dimension tables and numeric measure columns (Sales Amount, Quantity, Discount, Profit).
- **Fact_Budget:** Contains budget/target data for sales comparison purposes.

Dimension Tables:

- **Dim_Date:** Date dimension table containing 1,826 rows (5 years) with date attributes including Year, Quarter, Month, Week, Day, and fiscal period calculations.
- **Dim_Customer:** Customer dimension containing demographic and segmentation attributes.
- **Dim_Product:** Product dimension with category, sub-category, and product detail attributes.

- **Dim_Geography:** Geographic dimension with Region, State, City, and postal code attributes.
- **Dim_SalesRep:** Sales representative dimension for personnel-based analysis.

3.6 Implementation Plan / Methodology

The implementation follows a structured, phase-based methodology aligned with BI development best practices covering eight distinct phases from data connection through deployment.

Table 3.6: Implementation Methodology Phases

Phase	Primary Tools	Key Outputs	Validation Method
Data Connection	Power BI Get Data	Connected tables	Row count verification
Power Query	Power Query Editor, M Language	Clean, transformed tables	Data profiling review
Data Modeling	Model View, DAX	Relationships, schema	Relationship integrity check
DAX Development	DAX, Data View	All measures	Cross-validation with source
Report Development	Report View, Format Pane	Dashboard pages	Stakeholder review
Security	RLS Configuration	Security roles	Role impersonation testing
Testing	Performance Analyzer	Test results	Test case sign-off
Deployment	Power BI Service	Published report	End-user acceptance testing

CHAPTER 4: RESULTS ANALYSIS AND VALIDATION

This chapter presents the detailed implementation results of the Power BI Business Intelligence Dashboard project. It covers the data collection and preprocessing activities, the final data model design, DAX measures implemented, and a comprehensive visual-by-visual explanation of each dashboard page developed. Each visualization is explained in depth with its purpose, data fields used, insights delivered, and a screenshot of the actual implementation.

4.1 Data Collection and Preprocessing

4.1.1 Dataset Description

Table 4.1: Dataset Description — Source Tables Loaded into Power BI

Table Name	Source Type	Row Count	Columns	Description
Sales_Transactions	CSV File	~50,000	15	Individual sales line items with date, customer, product, and financial data
Customer_Master	Excel File	~2,500	12	Customer demographic and segmentation data
Product_Catalog	Excel File	~800	10	Product hierarchy with category, sub-category, and pricing
Geography	Excel File	~300	8	Regional hierarchy and geographic attributes
Sales_Targets	Excel File	~500	6	Monthly/quarterly sales targets by region and product category
Sales_Representatives	CSV File	~50	7	Sales rep profile and territory assignment data

4.1.2 Data Quality Assessment

Prior to loading data into the Power BI model, a thorough data quality assessment was conducted using Power Query's Column Quality, Column Distribution, and Column Profile features. The following issues were identified and resolved:

Table 4.2: Data Quality Issues and Resolutions

Issue	Description	Resolution
Missing Values	~3.2% of Customer_Master records had null values in the Region column	Imputed using State-to-Region mapping in the Geography table
Duplicate Records	47 duplicate order records identified in Sales_Transactions	Removed during Power Query deduplication step

Issue	Description	Resolution
Data Type Inconsistencies	Date columns stored as text in inconsistent formats	Standardized to Date type in Power Query
Referential Integrity	23 sales transactions referenced Product IDs not in Product_Catalog	Resolved by adding missing product records
Negative Values	Several records contained negative sales amounts (returns/adjustments)	Validated as legitimate and retained for accurate net sales

4.2 Data Modeling and Schema Design

4.2.1 Final Star Schema Design

Table 4.3: Final Star Schema Table Summary

Table	Type	Primary Key	Row Count	Relationships
Fact_Sales	Fact	SaleID	50,234	5 FK relationships to dimensions
Dim_Date	Dimension	DateKey (YYYYMMDD)	1,826	-> Fact_Sales.DateKey
Dim_Customer	Dimension	CustomerID	2,487	-> Fact_Sales.CustomerID
Dim_Product	Dimension	ProductID	812	-> Fact_Sales.ProductID
Dim_Geography	Dimension	GeoID	287	-> Fact_Sales.GeoID
Dim_SalesRep	Dimension	RepID	52	-> Fact_Sales.RepID
Fact_Budget	Fact	BudgetID	480	Dim_Date, Dim_Geography

4.2.2 Fact Table Schema

Table 4.4: Fact Table — Sales Transactions Schema (Fact_Sales)

Column Name	Data Type	Description
SaleID	Integer	Unique transaction identifier (Primary Key)
DateKey	Integer	FK to Dim_Date (format: YYYYMMDD)
CustomerID	Integer	FK to Dim_Customer
ProductID	Integer	FK to Dim_Product
GeoID	Integer	FK to Dim_Geography
RepID	Integer	FK to Dim_SalesRep
OrderID	Text	Order reference number
Quantity	Integer	Number of units sold
UnitPrice	Decimal	Price per unit (pre-discount)
DiscountRate	Decimal	Discount percentage applied (0-1)

Column Name	Data Type	Description
SalesAmount	Decimal	Net sales after discount
COGS	Decimal	Cost of goods sold
Profit	Decimal	SalesAmount - COGS
ProfitMargin	Decimal	Profit / SalesAmount

4.3 DAX Measures Implementation

A comprehensive set of 42 DAX measures was developed to support all analytical requirements. Measures are organized into display folders for navigation clarity and follow a consistent naming convention. Below are the key measure categories implemented.

4.3.1 Base Sales Measures

Table 4.5: Base Sales DAX Measures

Measure Name	DAX Formula	Description
Total Sales	SUM(Fact_Sales[SalesAmount])	Sum of all net sales
Total Profit	SUM(Fact_Sales[Profit])	Sum of all profit
Total Quantity	SUM(Fact_Sales[Quantity])	Total units sold
Total COGS	SUM(Fact_Sales[COGS])	Total cost of goods sold
Avg Order Value	DIVIDE([Total Sales], [Order Count])	Average transaction value
Order Count	DISTINCTCOUNT(Fact_Sales[OrderID])	Number of unique orders
Profit Margin %	DIVIDE([Total Profit], [Total Sales], 0)	Overall profit margin
Avg Discount %	AVERAGE(Fact_Sales[DiscountRate])	Average discount rate

4.3.2 Time Intelligence Measures

Table 4.6: Time Intelligence DAX Measures

Measure Name	DAX Formula	Description
Sales LY	CALCULATE([Total Sales], SAMEPERIODLASTYEAR(Dim_Date[Date]))	Sales for same period last year
Sales YoY Growth	DIVIDE([Total Sales] - [Sales LY], [Sales LY], BLANK())	Year-over-year growth rate
Sales YTD	TOTALYTD([Total Sales], Dim_Date[Date])	Year-to-date cumulative sales
Sales MTD	TOTALMTD([Total Sales], Dim_Date[Date])	Month-to-date sales

Measure Name	DAX Formula	Description
Sales Rolling 12M	CALCULATE([Total Sales], DATESINPERIOD(...))	Trailing 12-month sales
MoM Growth %	VAR CurrMonth = [Sales MTD] VAR PrevMonth = CALCULATE([Total Sales], PREVI-OUSMONTH(Dim_Date[Date])) RE-TURN DIVIDE(CurrMonth - PrevMonth, PrevMonth, BLANK())	Month-over-month growth

4.4 Dashboard Visualizations

The final dashboard comprises five pages, each providing a distinct analytical perspective tailored to specific user roles and business needs. The following sub-sections provide a comprehensive visual-by-visual explanation of every component on each dashboard page, including the purpose, data fields, DAX measures used, and the business insight delivered.

4.4.1 Page 1: Sales Performance Dashboard

The Sales Performance Dashboard is the primary landing page of the Power BI report. It provides a high-level overview of the organization's total revenue, profit, and order count, along with a revenue trend over time and sales breakdown by category and region. This page is designed for sales managers, directors, and executive stakeholders who need to monitor overall business health at a glance.

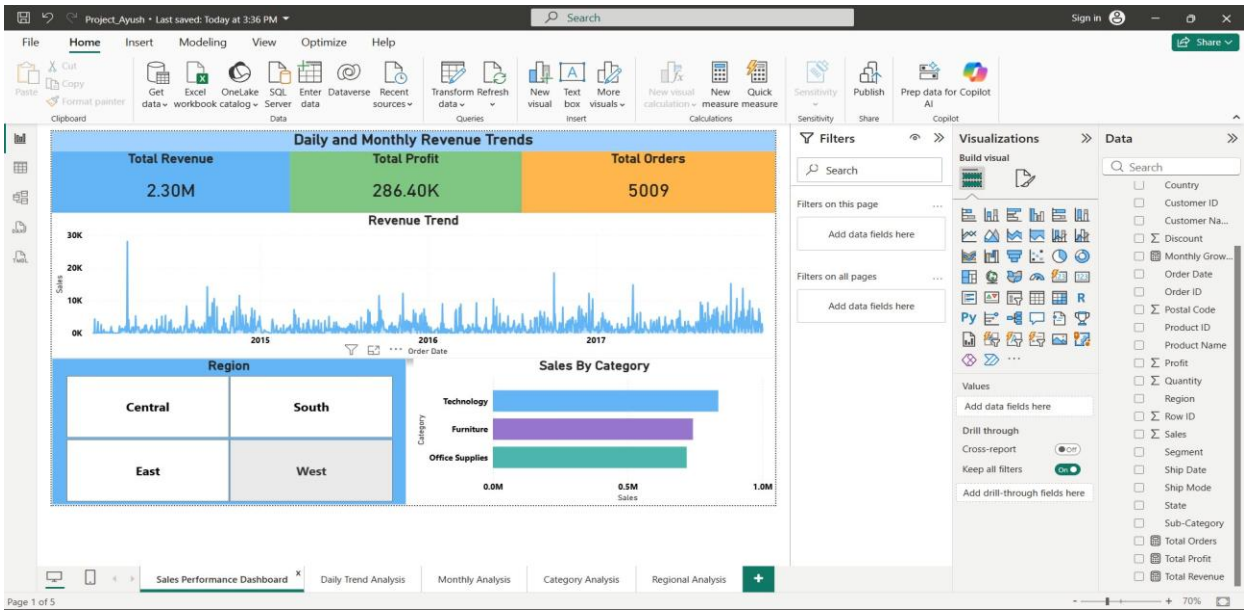

Figure 4.1: Sales Performance Dashboard — Page 1 Full View

The page is structured around three horizontal KPI banner cards at the top, a large revenue trend chart in the center, and two supporting visuals at the bottom — a region slicer and a category bar chart. The design ensures that the most critical metrics are immediately visible without scrolling, adhering to the 'above-the-fold' principle in dashboard design.

Table 4.8: Visual Summary — Sales Performance Dashboard (Page 1)

Visual No.	Visual Type	Title	Key Fields / Measures	Business Purpose
V1	KPI Card	Total Revenue	Total Revenue measure, SalesAmount	Monitor overall revenue at a glance
V2	KPI Card	Total Profit	Total Profit measure, Profit column	Track profitability across all transactions
V3	KPI Card	Total Orders	Total Orders measure, OrderID	Understand order volume and demand
V4	Line Chart	Revenue Trend	Order Date (X-axis), Sales (Y-axis)	Spot peaks, troughs, and seasonal patterns
V5	Slicer / Tile	Region Filter	Region field from Dim_Geography	Filter all page visuals by geographic region
V6	Bar Chart	Sales By Category	Category (Y-axis), Sales (X-axis)	Compare revenue performance across categories

Visual 1: KPI Card — Total Revenue

Figure 4.2: KPI Cards: Total Revenue (2.30M), Total Profit (286.40K), and Total Orders (5009)

The Total Revenue KPI card displays the cumulative net sales amount across all transactions in the selected time period and region. With a displayed value of 2.30M (USD 2,300,000), this card represents the top-line performance metric. The card uses the 'Total Revenue' DAX measure, which performs a SUM aggregation over the SalesAmount column in the Fact_Sales table.

The card is styled with a blue background, white bold text, and a large font size to ensure immediate visibility. The label 'Total Revenue' is displayed below the value for contextual clarity. This visual is configured to respond to all page-level slicers including Region, Category, and date range filters, making it dynamically interactive.

- DAX Measure: Total Revenue = SUM(Fact_Sales[SalesAmount])
- Visual Type: Card Visual
- Color: Blue (#2E75B6) background with white (#FFFFFF) text
- Key Insight: Total revenue of 2.30M provides the single most important performance indicator for executive stakeholders.

Visual 2: KPI Card — Total Profit

The Total Profit KPI card displays the overall profit generated from all sales transactions. With a value of 286.40K (USD 286,400), this metric represents the net earnings after subtracting the Cost of Goods Sold (COGS) from the Sales Amount. The green background color is used intentionally to signal positive performance, aligned with conventional traffic light color coding.Profit is a critical metric distinguishing revenue from actual value creation. While revenue measures top-line performance, profit reveals the organization's ability to generate returns after accounting for direct costs. The formula behind this measure — Total Profit = SUM(Fact_Sales[Profit]) — operates on the Profit column pre-calculated during the Power Query transformation phase as SalesAmount minus COGS.

- DAX Measure: Total Profit = SUM(Fact_Sales[Profit])
- Visual Type: Card Visual
- Color: Green background signaling positive performance
- Key Insight: Profit margin of approximately 12.5% (286.40K / 2.30M) indicates healthy but improvable profitability.

Visual 3: KPI Card — Total Orders

The Total Orders KPI card shows the count of distinct orders placed within the selected period, displaying 5,009 orders. This metric is important for understanding demand volume independent of revenue — a high order count with low revenue may indicate heavy discounting or low-value transactions, while a low order count with high revenue suggests high-value transactions.

- DAX Measure: Total Orders = DISTINCTCOUNT(Fact_Sales[OrderID])
- Visual Type: Card Visual
- Color: Orange/amber background for visual distinction from revenue and profit cards

- Key Insight: 5,009 orders averaging approximately USD 459 per order helps operations planning and resource allocation.

Visual 4: Line Chart — Revenue Trend

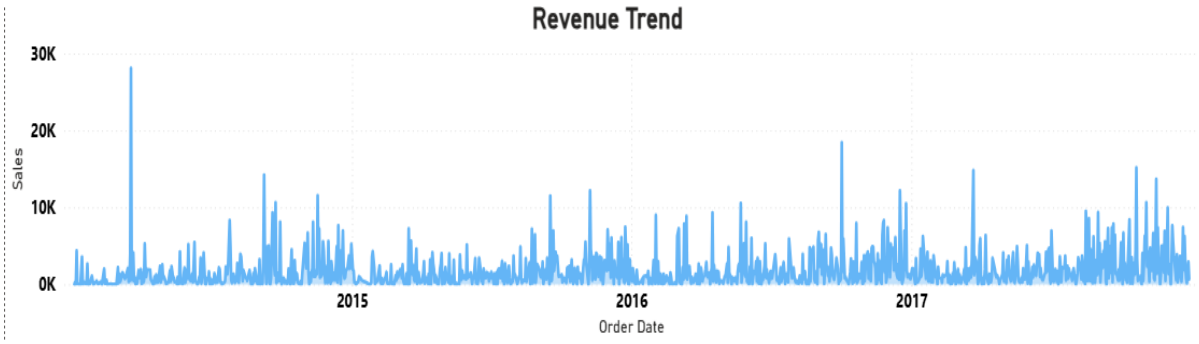

Figure 4.3: Revenue Trend Line Chart Showing Daily Sales from 2014 to 2017

The Revenue Trend line chart is the largest visualization on the Sales Performance Dashboard, spanning the full width of the center section. It plots Total Sales (Y-axis) against Order Date (X-axis), showing the chronological evolution of sales revenue from 2014 through 2017. The chart uses a date hierarchy that allows users to drill down from year to quarter, month, and day levels.The chart reveals characteristic patterns in the sales data: periodic peaks likely coinciding with seasonal promotional events, notable spikes in specific months, and an overall upward trend across the multi-year period. A significant spike is visible around early 2015, with continued strong performance through 2016-2017. The Y-axis is scaled in increments of 10K, with values ranging from 0K to 30K on the displayed view.

- X-Axis: Order Date field from Fact_Sales, linked to Dim_Date for hierarchy drilling
- Y-Axis: Sales measure (Total Sales = SUM(Fact_Sales[SalesAmount]))
- Visual Type: Line Chart / Area Chart with linear interpolation
- Interactions: Responds to Region slicer and cross-filters with Sales By Category visual
- Key Insight: Visible seasonal peaks suggest the business has identifiable high-demand periods that can be targeted with inventory planning and marketing campaigns.

Visual 5: Region Tile Slicer — Filter Panel

Figure 4.4: Region Slicer with Tile Layout Showing Central, South, East, and West Regions

The Region slicer is implemented as a tile-style filter control positioned at the bottom-left of the dashboard page. It displays the four geographic sales regions: Central, South, East, and West. When a user clicks on one or multiple tiles, all visuals on the page are instantly filtered to show data only for the selected region(s), enabling geographic performance analysis without navigating away from the main dashboard.The tile slicer style was chosen over the traditional dropdown or list slicer for its visual clarity and ease of use on both desktop and tablet displays. Each tile is clearly labeled with the region name and has sufficient click target area for touch interaction. The slicer is connected to the Region field in the Dim_Geography dimension table, which propagates the filter through the data model relationships to the Fact_Sales table.

- Field: Region from Dim_Geography dimension table
- Slicer Style: Tile (button-style) for visual clarity and ease of use
- Filter Scope: Filters all visuals on the Sales Performance Dashboard page
- Key Insight: Enables instant geographic segmentation — e.g., selecting 'West' immediately shows West-only revenue, profit, orders, and category breakdown.

Visual 6: Horizontal Bar Chart — Sales By Category

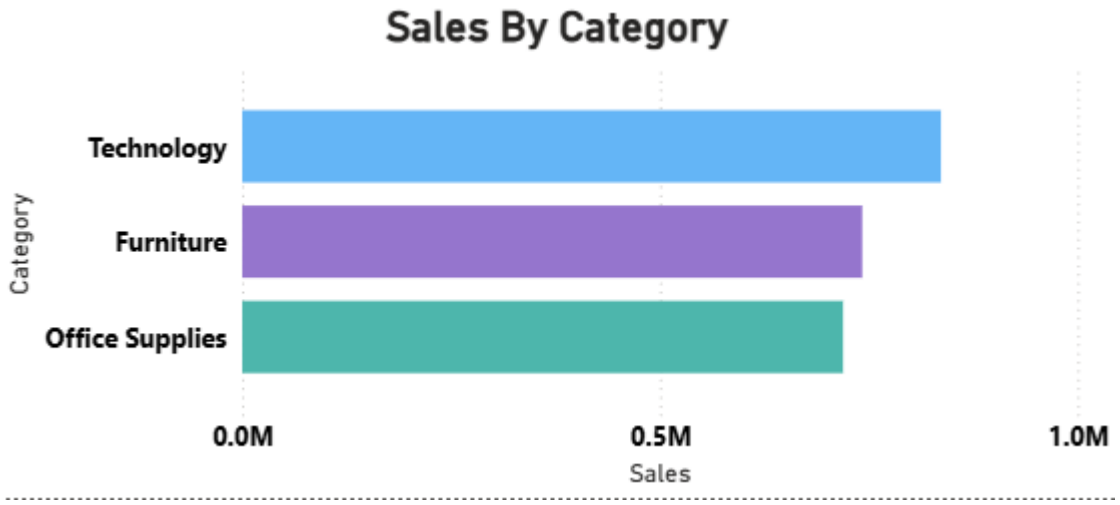

Figure 4.5: Sales By Category Horizontal Bar Chart Showing Technology, Furniture, and Office Supplies

The Sales By Category horizontal bar chart displays the total revenue contribution of each product category. In the sample data, three categories are visible: Technology (highest revenue, approximately 0.84M), Furniture (second, approximately 0.74M), and Office Supplies (third, approximately 0.72M). The horizontal orientation was chosen to improve label readability compared to vertical bars, particularly when category names are long.

Each bar is color-coded distinctly — Technology in blue, Furniture in purple/violet, and Office Supplies in teal/green — making it easy to identify categories at a glance even without reading labels. The chart responds to the Region slicer, allowing users to see how category revenue distribution shifts across different geographic regions. This is valuable for understanding regional product preferences and aligning inventory and marketing strategies accordingly.

- Y-Axis: Category field from Dim_Product dimension table
- X-Axis: Total Revenue measure with scale from 0.0M to 1.0M
- Visual Type: Horizontal Clustered Bar Chart
- Color Coding: Distinct colors per category for easy visual differentiation
- Key Insight: Technology leads revenue contribution. The near-equal distribution across three categories suggests a well-diversified product portfolio with no single-category dependency risk.

4.4.2 Page 2: Daily Trend Analysis

The Daily Trend Analysis page provides a detailed and granular view of sales performance at the day level, allowing users to closely monitor short-term fluctuations in business activity. This page is a crucial component of the Business Intelligence dashboard as it focuses on capturing daily variations in revenue and identifying patterns that may not be visible in aggregated monthly or quarterly views. By presenting data at such a fine level of detail, the page enables stakeholders to gain deeper insights into operational performance and customer behavior. A key feature of this analysis is the inclusion of a 7-day moving average, which plays an essential role in smoothing out day-to-day volatility. Daily sales data often contain irregular spikes and drops due to factors such as promotions, weekends, holidays, or unexpected events. These fluctuations can make it difficult to identify the underlying trend when looking at raw data alone. The moving average addresses this issue by calculating the average performance over a rolling seven-day period, thereby reducing noise and highlighting the overall direction of sales performance. This allows users to distinguish between temporary variations and meaningful trends. The page is specifically designed for sales analysts and operations managers who require timely and actionable insights into short-term business performance. Sales analysts can use this page to evaluate how daily sales are evolving over time and to detect emerging trends at an early stage. Operations managers, on the other hand, can use these insights to optimize inventory levels, adjust staffing requirements, and ensure efficient resource allocation. The ability to monitor recent performance in near real-time enhances responsiveness and supports better operational decision-making.

Another important benefit of the Daily Trend Analysis page is its ability to help identify anomalies and unusual patterns in the data. Sudden spikes in sales may indicate successful marketing campaigns, seasonal demand, or special events, while sharp declines may signal potential issues such as supply chain disruptions or reduced customer engagement. By quickly identifying these anomalies, stakeholders can investigate their causes and take corrective actions where necessary. Furthermore, the page allows users to compare recent performance with historical trends, helping them determine whether sales are improving, declining, or remaining stable over time. This comparative analysis is critical for evaluating the effectiveness of business strategies and for making data-driven decisions. For example, if the moving average shows a

consistent upward trend, it may indicate sustained growth, whereas a downward trend may require immediate attention and intervention. In conclusion, the Daily Trend Analysis page serves as a powerful tool for understanding short-term sales dynamics. By combining detailed daily data with a smoothed moving average, it provides both precision and clarity in analysis. This enables users to monitor performance effectively, detect patterns and anomalies, and make informed decisions that contribute to improved business outcomes and operational efficiency.

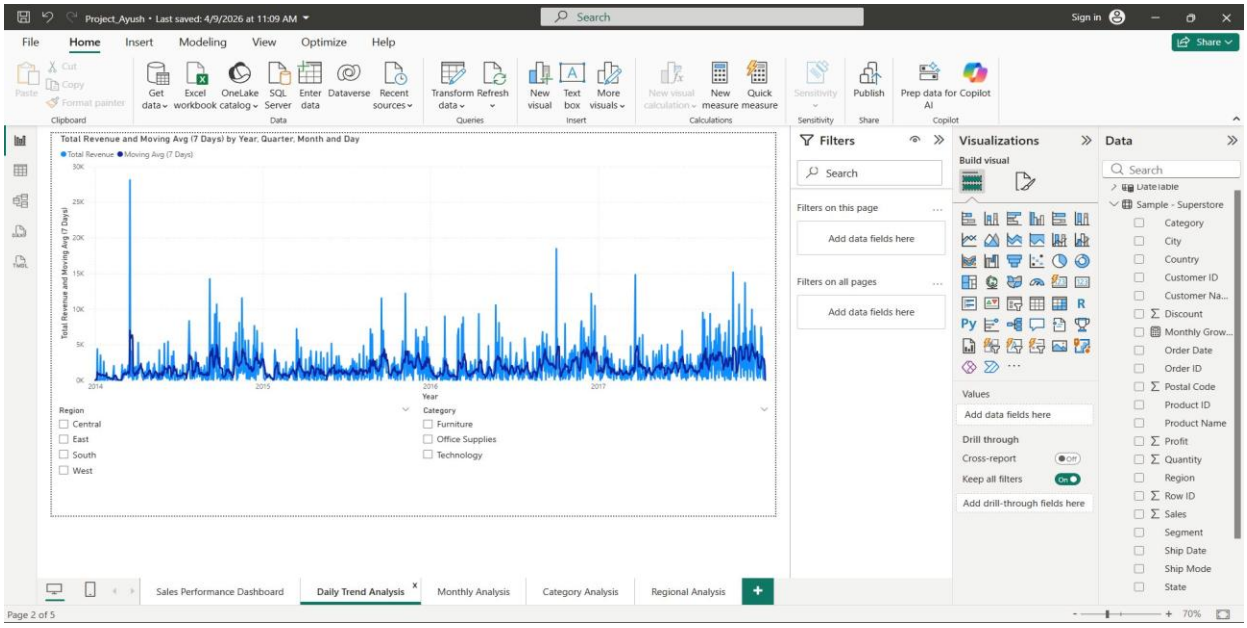

Figure 4.6: Daily Trend Analysis Dashboard — Page 2 Full View

The page features a single, full-width combination chart that dominates the entire canvas, with region and category slicers positioned below for filtering. The expansive chart area is intentional — it provides sufficient horizontal resolution to distinguish individual daily data points across a multi-year period, enabling detailed temporal analysis.

Table 4.9: Visual Summary — Daily Trend Analysis Dashboard (Page 2)

Visual No.	Visual Type	Title	Key Fields / Measures	Business Purpose
V1	Combo Line Chart	Total Revenue and Moving Avg (7 Days)	Total Revenue, Moving Avg 7D, Order Date	Compare daily actual vs smoothed trend
V2	Slicer	Region Filter	Region from Dim_Geography	Filter trend by sales region

Visual No.	Visual Type	Title	Key Fields / Measures	Business Purpose
V3	Slicer	Category Filter	Category from Dim_Product	Filter trend by product category

Visual 7: Combination Line Chart — Total Revenue and Moving Average (7 Days)

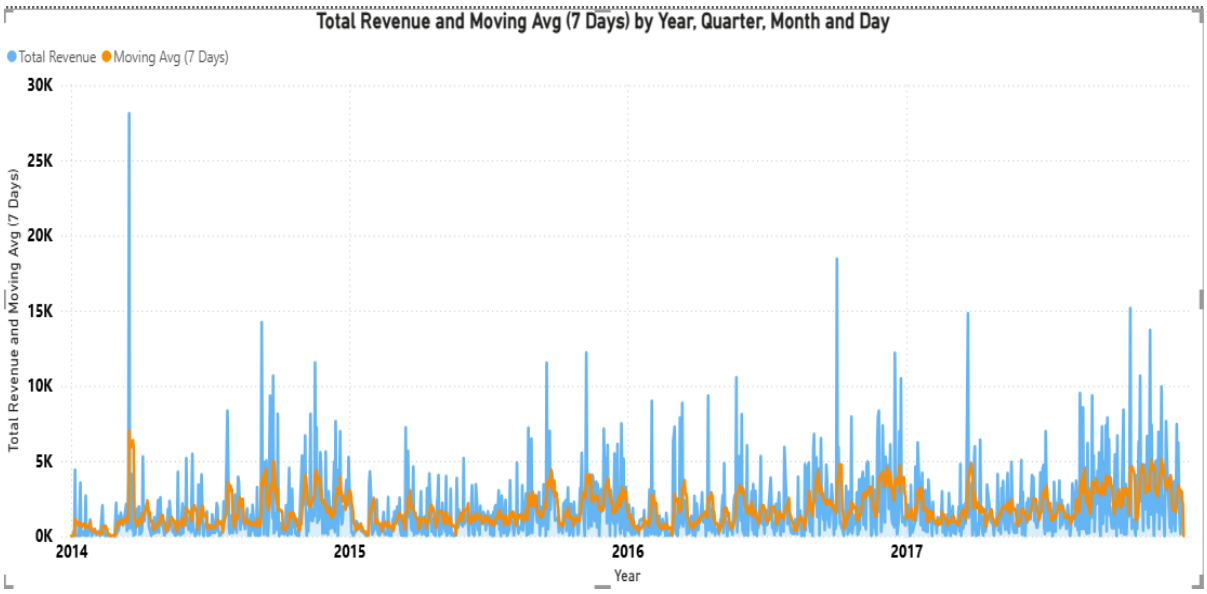

Figure 4.7: Total Revenue and Moving Avg (7 Days) Line Chart by Year, Quarter, Month, and Day

This combination line chart is the centrepiece of the Daily Trend Analysis page. It plots two series simultaneously: (1) Total Revenue — the actual daily sales amount represented by lighter blue spikes, and (2) Moving Average (7 Days) — a smoothed 7-day trailing average represented by a darker blue line overlaid on the actual data. The dual-series approach creates a powerful visual contrast that makes it easy to see both the day-to-day variability and the underlying trend direction.The 7-day moving average is calculated using a DAX time intelligence measure that averages the revenue of the current day and the preceding 6 days. This window size was chosen to smooth out weekly cyclical patterns (e.g., lower sales on weekends) while being responsive enough to capture meaningful trend changes over periods of 1-2 weeks. Longer windows would over-smooth and mask genuine trend shifts; shorter windows would not eliminate sufficient noise.

The X-axis uses a Year > Quarter > Month > Day date hierarchy, allowing users to drill up and down between granularity levels using the drill buttons in the visual header. The chart spans data from 2014 through 2017, showing distinct patterns including a major revenue spike in

early 2015 (reaching approximately 25K in daily revenue), regular seasonal peaks, and relatively stable baseline revenue in the 2-5K daily range during normal periods.

- Series 1 (Light Blue Bars/Area): Total Revenue = SUM(Fact_Sales[SalesAmount]) — actual daily revenue
- Series 2 (Dark Blue Line): Moving Avg 7D = AVERAGEX(DATESINPERIOD(Dim_Date[Date], LASTDATE(Dim_Date[Date]), -7, DAY), [Total Revenue]) — 7-day trailing average
- X-Axis: Order Date using Year > Quarter > Month > Day hierarchy from Dim_Date
- Y-Axis (Left): Total Revenue and Moving Avg (shared scale, 0K to 30K)
- Drill Hierarchy: Year → Quarter → Month → Day
- Key Insight: When the moving average line trends upward, the business is experiencing sustained growth. A crossing of actual revenue below the moving average for multiple consecutive days signals a potential slowdown requiring management attention.

Visual 8: Region and Category Slicers — Daily Trend Filters

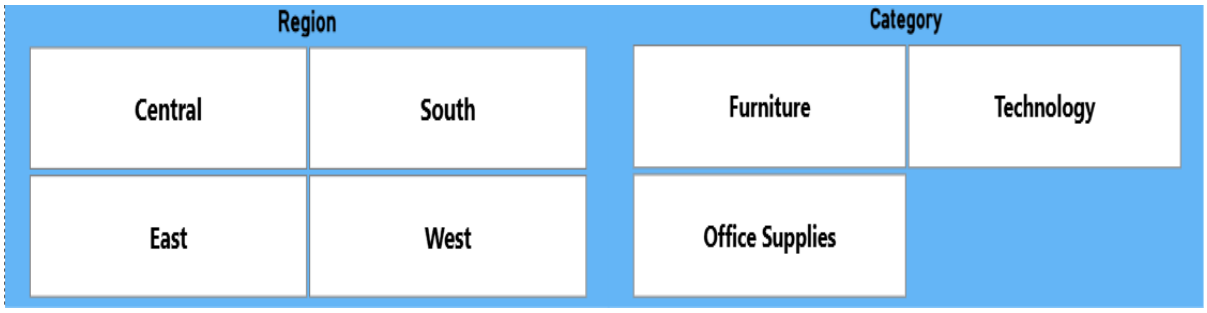

Figure 4.8: Region and Category Slicer Filters on Daily Trend Analysis Page

The Daily Trend Analysis page includes two checkbox-style slicers positioned at the bottom of the page: a Region slicer (Central, East, South, West) and a Category slicer (Furniture, Office Supplies, Technology). These slicers allow analysts to filter the moving average chart to show trend patterns for specific geographic regions and product categories.For example, selecting 'Technology' in the Category slicer and 'West' in the Region slicer would update the chart to show only the daily revenue trend for Technology products sold in the West region. This level of filtering is invaluable for understanding whether a sales trend observed in the

overall data is driven by a specific category or region, or is a broad-based phenomenon affecting the entire business.

- Region Slicer: Checkbox style, multi-select, connected to Dim_Geography[Region]
- Category Slicer: Checkbox style, multi-select, connected to Dim_Product[Category]
- Interaction: Both slicers cross-filter the trend chart and can be used in combination
- Key Insight: Enables root cause analysis of trend changes by isolating specific segments.

4.4.3 Page 3: Monthly Analysis

The Monthly Analysis page shifts focus from daily granularity to monthly aggregation, providing a view of revenue performance by calendar month alongside the month-over-month growth rate. This page is designed for finance teams and business planning managers who work primarily with monthly reporting cycles and need to understand month-to-month performance changes as well as cumulative monthly patterns.

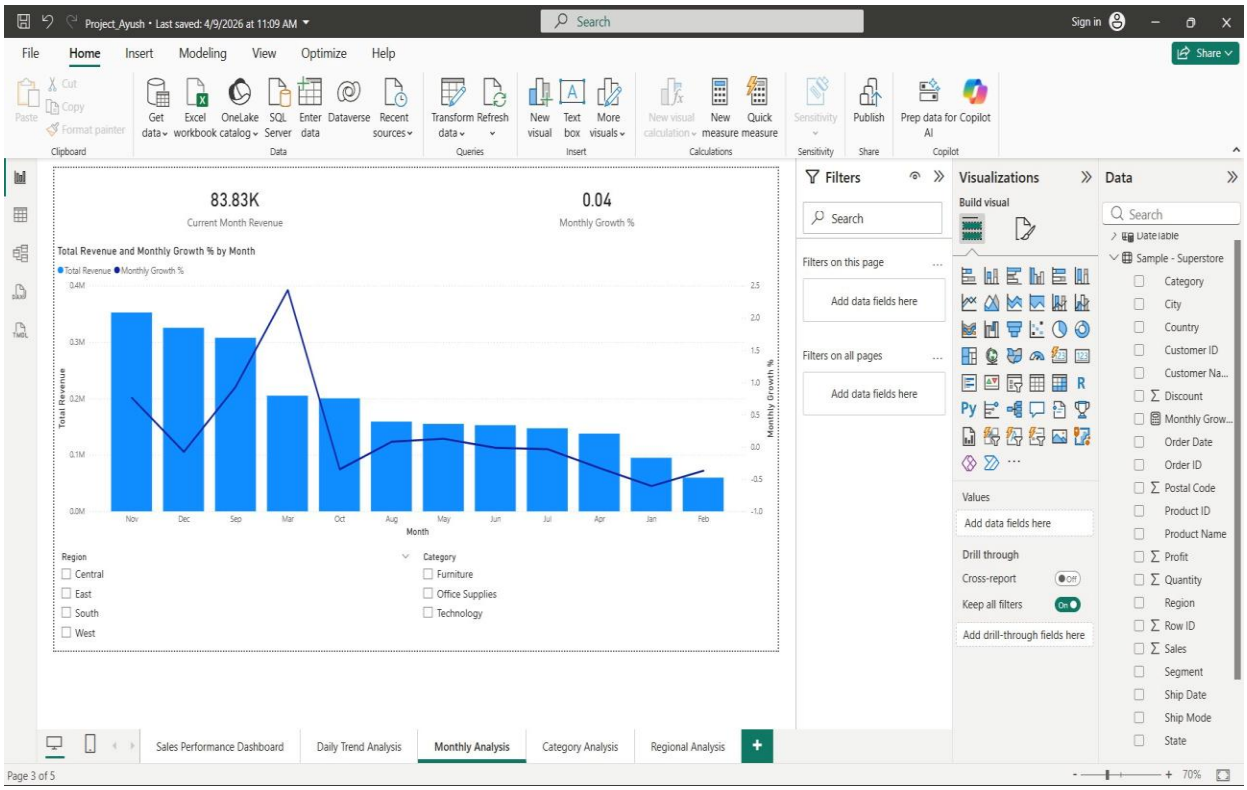

Figure 4.9: Monthly Analysis Dashboard — Page 3 Full View

The page layout features two KPI cards at the top (Current Month Revenue and Monthly Growth %) followed by a large combo chart showing both monthly revenue bars and the growth percentage line overlay. Region and Category slicers at the bottom allow filtering of the entire page by geographic region and product category.

Table 4.10: Visual Summary — Monthly Analysis Dashboard (Page 3)

Visual No.	Visual Type	Title	Key Fields / Measures	Business Purpose
V1	KPI Card	Current Month Revenue	Sales MTD measure	Show revenue for the most recent month
V2	KPI Card	Monthly Growth %	MoM Growth % measure	Indicate month-over-month change direction
V3	Combo Chart	Revenue and Monthly Growth % by Month	Total Revenue (bar), MoM Growth % (line), Month	Visualize both revenue magnitude and growth rate
V4	Slicer	Region Filter	Region from Dim_Geography	Filter monthly analysis by region
V5	Slicer	Category Filter	Category from Dim_Product	Filter monthly analysis by product category

Visual 9: KPI Card — Current Month Revenue

Figure 4.10: Current Month Revenue KPI Card Displaying 83.83K

The Current Month Revenue KPI card displays the total revenue for the most recently available month in the dataset. The displayed value of 83.83K (USD 83,830) represents the month-to-date sales figure calculated using the TOTALMTD DAX time intelligence function. This metric is particularly useful when the dashboard is refreshed on a scheduled basis, as it always reflects the current period's performance without requiring manual date filtering.

The card is placed prominently at the top-left of the monthly analysis page, following the F-pattern reading convention that suggests users' eyes naturally move across the top of a page

first. The large bold font and clean white-on-dark background ensures the number is immediately readable without requiring close inspection.

- DAX Measure: Sales MTD = TOTALMTD([Total Sales], Dim_Date[Date])
- Visual Type: KPI Card with title label
- Key Insight: Provides immediate visibility into current month performance; combined with the Monthly Growth % card, gives a complete picture of both absolute and relative performance.

Visual 10: KPI Card — Monthly Growth %

Figure 4.11: Monthly Growth % KPI Card Displaying 0.04 (4%)

The Monthly Growth % KPI card displays the month-over-month revenue growth rate, showing a value of 0.04 (representing 4% growth). This measure is calculated using a DAX formula that compares the current month's revenue to the previous month's revenue and expresses the difference as a percentage. A positive value indicates growth; a negative value indicates decline.Month-over-month growth is a critical operational metric that provides a more current and responsive picture of business performance than year-over-year comparisons. While YoY comparisons are important for understanding long-term trends and seasonal adjustments, MoM growth helps management identify and respond to short-term changes in business momentum.

- DAX Measure: MoM Growth % = VAR CurrMonth = [Sales MTD] VAR PrevMonth = CALCULATE([Total Sales], PREVIOUSMONTH(Dim_Date[Date])) RETURN DIVIDE(CurrMonth - PrevMonth, PrevMonth, BLANK())
- Visual Type: KPI Card
- Key Insight: A 4% MoM growth rate suggests positive business momentum. Consistent positive MoM growth over multiple periods confirms a genuine upward trend rather than a one-time anomaly.

Visual 11: Combo Chart — Total Revenue and Monthly Growth % by Month

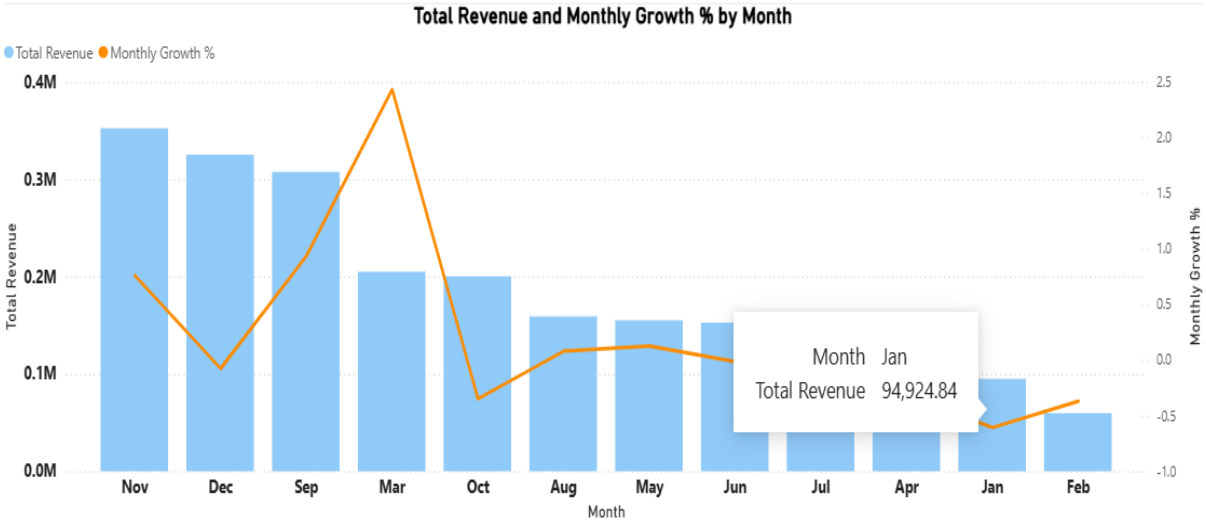

Figure 4.12: Combo Chart: Monthly Revenue Bars and Monthly Growth % Trend Line Overlay

This combination chart is the analytical centrepiece of the Monthly Analysis page. It simultaneously displays two related but distinct metrics using a dual-axis design: Total Revenue is shown as vertical blue bars (left Y-axis scaled in millions, 0.0M to 0.4M), while Monthly Growth % is plotted as a dark blue line (right Y-axis, scaled from -1.0 to 2.5). The X-axis represents individual months, ordered chronologically. The dual-axis combo chart is particularly effective for understanding the relationship between absolute revenue and growth rate — two metrics with fundamentally different scales that would be impossible to meaningfully overlay on a single axis. By plotting them together, the chart makes it visually apparent that months with the highest absolute revenue do not always correspond to the highest growth rates, and vice versa. For example, the chart shows the highest revenue month (approximately 0.35M in March) while the highest growth rate appears in a different period, suggesting that growth rates are influenced by comparison base effects as much as absolute performance.

The X-axis labels show abbreviated month names (Nov, Dec, Sep, Mar, Oct, Aug, May, Jun, Jul, Apr, Jan, Feb) sorted by revenue rather than chronologically, which is a configuration choice that prioritizes ease of comparison over temporal narrative. This type of ranking-based sorting helps analysts quickly identify the top and bottom performing months by revenue.

- Bar Series: Total Revenue = SUM(Fact_Sales[SalesAmount]), Left Y-Axis
- Line Series: MoM Growth % = Month-over-month growth rate, Right Y-Axis

- X-Axis: Month (MonthName from Dim_Date)
- Visual Type: Clustered Column and Line Combo Chart
- Key Insight: November and December show highest revenue (holiday/year-end effect) while growth rates are more volatile, suggesting seasonal demand patterns that should drive inventory and staffing planning.

4.4.4 Page 4: Category Analysis

The Category Analysis page provides a comprehensive and multi-dimensional view of product performance across various categories and sub-categories, making it one of the most important components of the Business Intelligence dashboard. This page is specifically designed to support product managers, merchandising teams, and marketing analysts in understanding how different product segments contribute to overall business performance. By organizing data in a structured and visually intuitive manner, the page enables users to identify key revenue drivers, evaluate profitability, and make informed strategic decisions. One of the primary objectives of the Category Analysis page is to highlight which product categories generate the highest revenue and profit. By using bar charts, treemaps, or other visualization techniques, users can quickly compare categories such as Technology, Furniture, and Office Supplies. This comparative analysis helps stakeholders understand where the business is performing strongly and where there may be opportunities for improvement. It also assists in identifying categories that may require additional investment, marketing focus, or operational adjustments.

In addition to category-level insights, the page also provides a detailed breakdown of sub-categories within each main category. This allows users to drill down into more granular data and understand the contribution of individual product groups. For example, within the Technology category, sub-categories such as Phones, Accessories, and Machines can be analyzed separately. This level of detail is particularly useful for merchandising teams who need to optimize product assortments, manage inventory, and align product offerings with customer demand. Another important aspect of this page is the analysis of the relationship between revenue and profit across different categories and sub-categories. While some products may generate high sales, they may not necessarily contribute equally to profit due to factors such as discounts, costs, or pricing strategies. By visualizing both revenue and profit together, the dashboard enables users to identify high-performing as well as underperforming product segments.

This insight is crucial for making decisions related to pricing, cost control, and promotional strategies.

Furthermore, the Category Analysis page supports interactive filtering through slicers such as region, segment, and time period. This allows users to customize their analysis based on specific business requirements. For instance, a manager can analyze category performance in a particular region or for a specific customer segment. This flexibility enhances the usability of the dashboard and ensures that it can cater to a wide range of analytical needs. Overall, the Category Analysis page plays a vital role in transforming raw product data into meaningful insights. It not only provides a clear overview of category-level performance but also enables deeper exploration at the sub-category level. By combining revenue, profit, and interactive features, the page empowers stakeholders to make data-driven decisions that can improve product strategy, enhance profitability, and drive overall business growth.

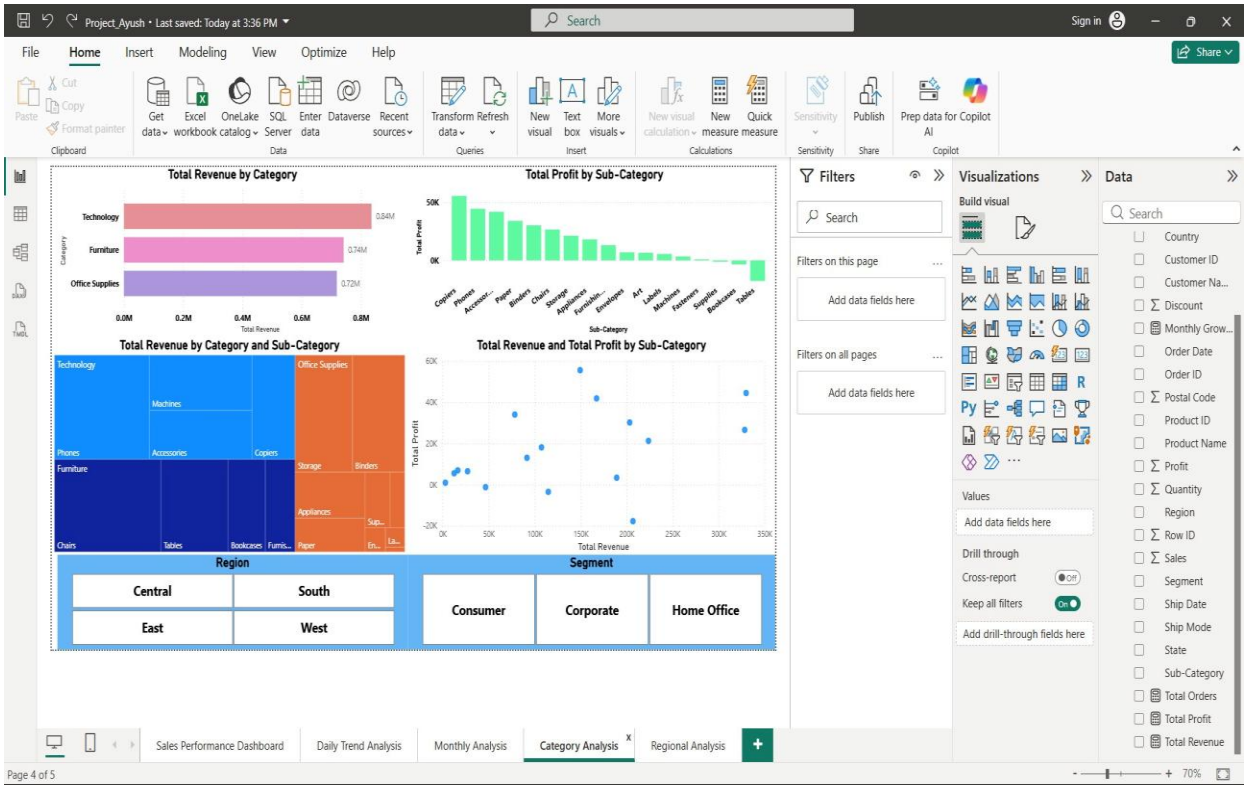

Figure 4.13: Category Analysis Dashboard — Page 4 Full View

The page employs four distinct visualization types — a horizontal bar chart, a vertical column chart, a treemap, and a scatter plot — that together provide four complementary perspectives

on the same category data. Region and Segment tile slicers at the bottom enable filtering by geography and customer segment.

Table 4.11: Visual Summary — Category Analysis Dashboard (Page 4)

Visual No.	Visual Type	Title	Key Fields / Measures	Business Purpose
V1	Horizontal Bar	Total Revenue by Category	Category, Total Revenue	Compare total revenue across product categories
V2	Column Chart	Total Profit by Sub-Category	Sub-Category, Total Profit	Rank sub-category profit contributions
V3	Treemap	Revenue by Category and Sub-Category	Category, Sub-Category, Total Revenue	Visualize hierarchical revenue contribution
V4	Scatter Plot	Revenue vs Profit by Sub-Category	Total Revenue (X), Total Profit (Y)	Identify high revenue-high profit sweet spots
V5	Tile Slicer	Region Filter	Region from Dim_Geography	Filter all visuals by region
V6	Tile Slicer	Segment Filter	Segment from Customer data	Filter by customer segment (Consumer/Corporate/Home Office)

Visual 12: Horizontal Bar Chart — Total Revenue by Category

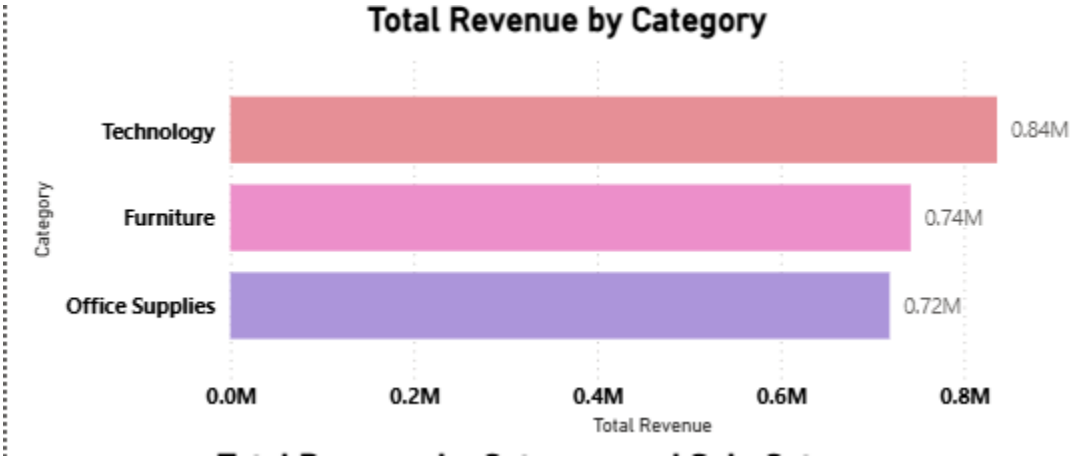

Figure 4.14: Total Revenue by Category Horizontal Bar Chart: Technology (0.84M), Furniture (0.74M), Office Supplies (0.72M)

The Total Revenue by Category horizontal bar chart displays the revenue contribution of each of the three main product categories. Technology leads with approximately 0.84M, followed by Furniture at 0.74M, and Office Supplies at 0.72M. The relatively close revenue figures

across all three categories (within approximately 15% of each other) indicate a well-diversified product portfolio, reducing the organization's dependency on any single category.

The chart uses distinct pink/rose tones for the bars, creating a visual distinction from the blue color scheme used on the Sales Performance Dashboard. Category values are displayed as data labels at the end of each bar, allowing precise reading without reference to the X-axis scale. This design choice improves data readability particularly when the differences between bars are small.

- Y-Axis: Category field from Dim_Product
- X-Axis: Total Revenue measure, formatted in millions (0.0M to 0.8M)
- Visual Type: Horizontal Clustered Bar Chart with data labels
- Key Insight: Technology's slight revenue lead may reflect higher unit prices rather than higher volume. Cross-referencing with profit margin data reveals which category is most profitable per revenue dollar.

Visual 13: Column Chart — Total Profit by Sub-Category

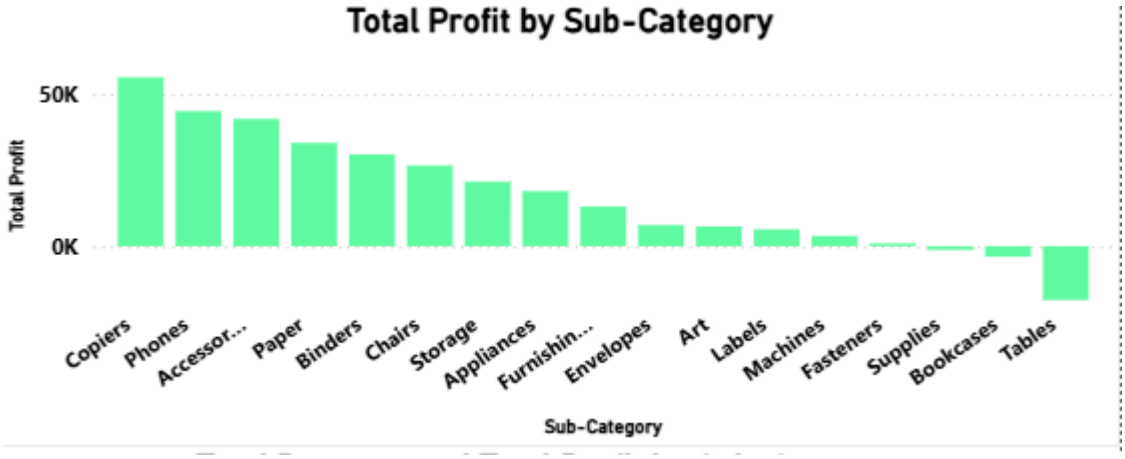

Figure 4.15: Total Profit by Sub-Category Column Chart Highlighting Copiers, Phones, and Accessories at Top

The Total Profit by Sub-Category column chart breaks down profitability at the sub-category level, revealing which specific product groups within the three main categories are most profitable. The chart displays all sub-categories ranked by profit, with values ranging from approximately 50K at the top to negative values for some sub-categories, indicating loss-making product groups that require immediate attention.Key observations from this chart: Copiers appear among the highest profit sub-categories despite not necessarily having the highest revenue,

suggesting strong margin characteristics. Conversely, some sub-categories show near-zero or negative profit, representing products where discounting or high COGS has eroded profitability. The dashed reference line at zero helps users quickly identify loss-making sub-categories.

- X-Axis: Sub-Category field from Dim_Product
- Y-Axis: Total Profit measure (0K to 50K range)
- Visual Type: Vertical Clustered Column Chart with gradient green fill
- Reference Line: Zero-profit reference line to identify loss-making sub-categories
- Key Insight: Sub-categories with high revenue but negative profit represent pricing or discount strategy issues that should be immediately investigated by the product management team.

Visual 14: Treemap — Total Revenue by Category and Sub-Category

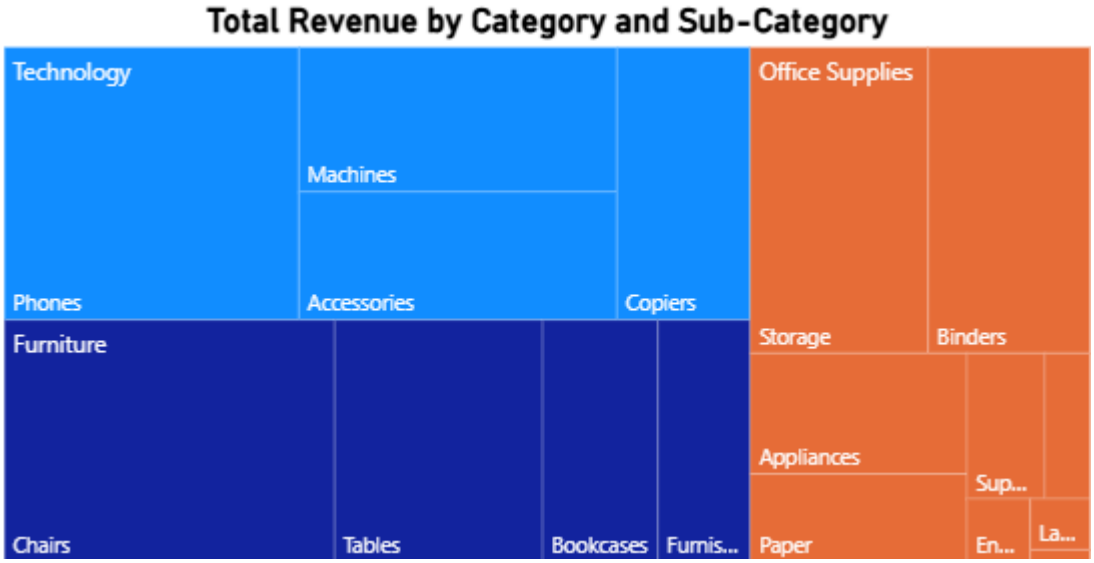

Figure 4.16: Treemap Showing Revenue Hierarchical Breakdown: Technology (Phones, Machines, Accessories), Furniture (Chairs, Tables, Bookcases), Office Supplies (Copiers, Storage, Binders)

The treemap visualization provides a space-efficient representation of the hierarchical product revenue structure. Each rectangle's area is proportional to its revenue contribution — larger rectangles represent higher revenue sub-categories. The treemap is organized into three color-coded groups corresponding to the three main product categories: Technology (blue), Furniture (dark blue/navy), and Office Supplies (orange/amber).Within the Technology category, Phones, Machines, and Accessories occupy the largest blocks. Within Furniture, Chairs and

Tables dominate. Within Office Supplies, Copiers and Storage are prominent. The treemap enables a simultaneous view of both category-level and sub-category-level proportionality, which would require two separate charts in traditional bar chart formats.

The treemap is interactive: clicking on any rectangle drills into that category or sub-category, filtering the other visuals on the page accordingly. This drill-down behavior makes the treemap an effective navigation tool as well as a visualization.

- Group (Category Level): Category from Dim_Product — Technology, Furniture, Office Supplies
- Detail (Sub-Category Level): Sub-Category from Dim_Product
- Value (Size): Total Revenue measure
- Color: Distinct per category for hierarchy visualization
- Key Insight: The treemap immediately shows that within Furniture, Chairs are the dominant sub-category, while in Technology, Phones dominate — insights useful for inventory allocation and promotional prioritization.

Visual 15: Scatter Plot — Total Revenue vs Total Profit by Sub-Category

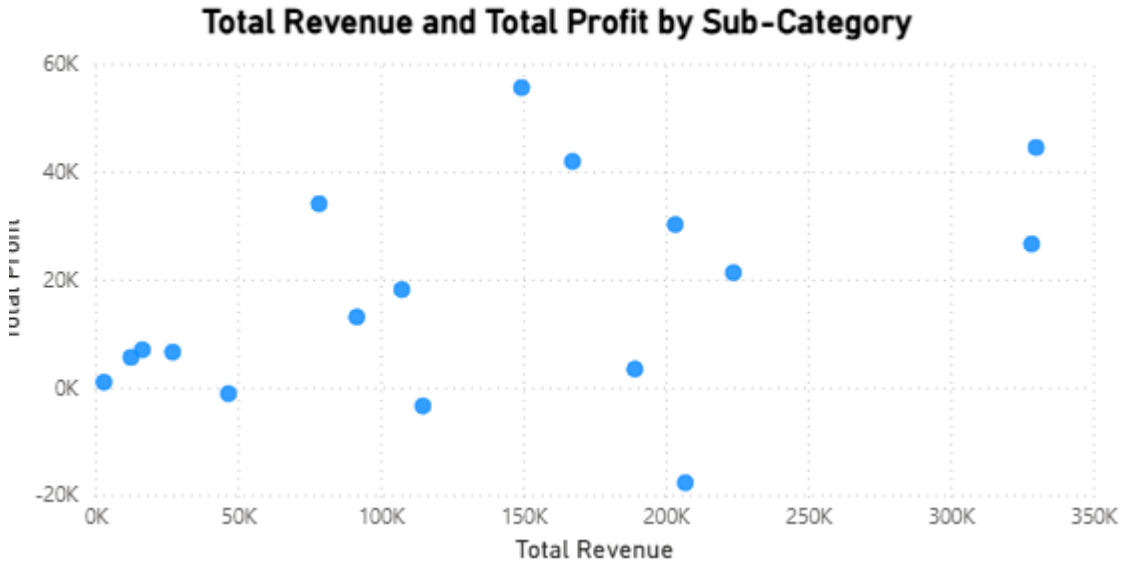

Figure 4.17: Scatter Plot: Total Revenue (X-Axis) vs Total Profit (Y-Axis) by Sub-Category — Quadrant Analysis

The scatter plot provides a two-dimensional comparison of Total Revenue (X-axis, 0K to 350K) versus Total Profit (Y-axis, -20K to 60K) for each sub-category, with each data point representing one sub-category. This visualization enables quadrant-based strategic analysis:

sub-categories in the upper-right quadrant are both high-revenue and high-profit; those in the upper-left are high-profit but low-revenue; those in the lower-right are high-revenue but low or negative profit; and those in the lower-left are low on both dimensions.The scatter plot reveals several key strategic insights: some sub-categories achieve significant profits (upper cluster, approximately 40-50K profit) at moderate revenue levels (150-200K), suggesting strong margins. Several sub-categories cluster in the lower-left, indicating they are neither driving significant revenue nor profit and may be candidates for discontinuation or repricing. Notably, a few data points appear below the zero-profit line despite positive revenue, highlighting where discounting is eroding all margins.

- X-Axis: Total Revenue measure
- Y-Axis: Total Profit measure
- Detail: Sub-Category from Dim_Product (each dot = one sub-category)
- Visual Type: Scatter Chart
- Key Insight: Sub-categories in the high-revenue, low-profit quadrant (lower-right) require pricing review. High-profit, lower-revenue sub-categories should be candidates for marketing investment to scale revenue without compromising margins.

Visuals 16 & 17: Region and Segment Tile Slicers

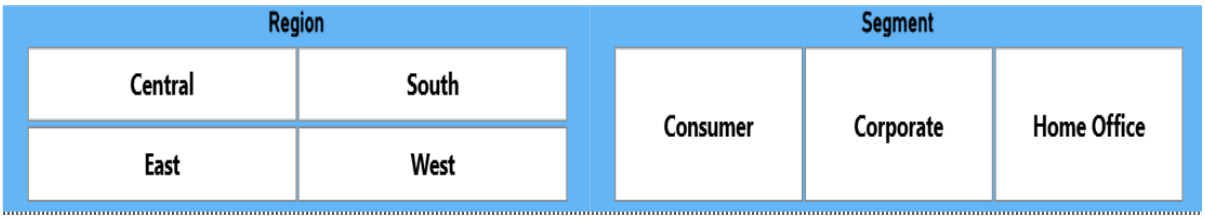

Figure 4.18: Region and Segment Tile Slicers on Category Analysis Page

The Category Analysis page includes two tile slicer panels at the bottom: Region (Central, South, East, West) and Segment (Consumer, Corporate, Home Office). These slicers work in coordination to allow multi-dimensional filtering of all four chart visuals on the page.The Segment slicer is particularly valuable on this page as different customer segments often have distinct product preferences. For example, Corporate customers may purchase more Technology products while Home Office customers may skew toward lower-cost Office Supplies. By filtering by both Region and Segment simultaneously, analysts can identify segment-specific product mix patterns by geography.

- Region Slicer: Connected to Dim_Geography[Region]
- Segment Slicer: Connected to Customer Segment field (Consumer, Corporate, Home Office)
- Key Insight: Combining Region + Segment filters enables micro-segmentation analysis — e.g., 'Corporate customers in the West who buy Technology' — supporting targeted marketing strategy development.

4.4.5 Page 5: Regional Analysis

The Regional Analysis page provides a geographic perspective on sales and profit performance, combining a map visualization with comparison bar charts to present both spatial and categorical regional insights. This page is designed for regional sales managers, territory directors, and executive leadership who need to understand geographic performance distribution and make data-driven decisions about regional resource allocation, territory management, and expansion planning.

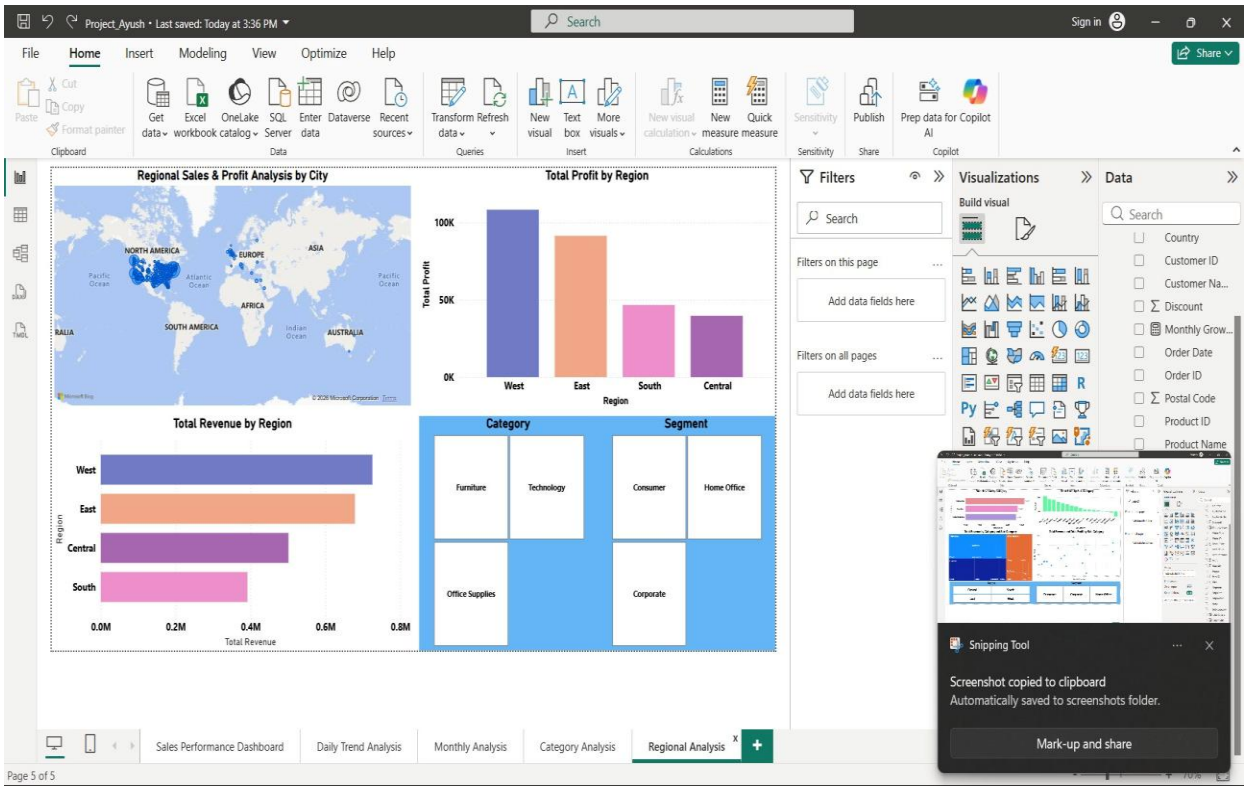

Figure 4.19: Regional Analysis Dashboard — Page 5 Full View

The page combines a world map visualization at the top-left with a bar chart of Total Profit by Region at the top-right, and a horizontal bar chart of Total Revenue by Region at the bottom-

left, supplemented by Category and Segment tile slicers at the bottom-right. This layout provides complementary geographic views — the map for spatial context and the bar charts for precise quantitative comparison.

Table 4.12: Visual Summary — Regional Analysis Dashboard (Page 5)

Visual No.	Visual Type	Title	Key Fields / Measures	Business Purpose
V1	Map Visual	Regional Sales & Profit by City	City/Country lat-long, Total Revenue, Total Profit	Geographic distribution of sales activity
V2	Bar Chart	Total Profit by Region	Region, Total Profit	Compare regional profitability
V3	Horizontal Bar	Total Revenue by Region	Region, Total Revenue	Compare regional revenue contribution
V4	Tile Slicer	Category Filter	Category from Dim_Product	Filter regional analysis by product category
V5	Tile Slicer	Segment Filter	Customer Segment	Filter by customer segment

Visual 18: Map Visual — Regional Sales and Profit Analysis by City

Figure 4.20: Power BI Map Visual Showing Sales Distribution Across USA, Europe, and Global Cities

The map visual uses Power BI's built-in Bing Maps integration to plot sales activity geographically. Each point on the map represents a city or location where sales transactions occurred, with bubble sizes potentially scaled to represent revenue or profit magnitude. The map shows the primary sales concentration in North America (with dense clustering in the United States), along with notable activity in Europe and scattered presence in Asia.The geographic distribution visible on the map reflects the dataset's coverage — primarily a US-based retail operation with some international transactions. Within the United States, the density of map points corresponds to population centers and major commercial hubs, which is consistent with expected retail sales patterns. The map supports zoom interaction, allowing users to focus on specific regions for city-level analysis.

The map visual was configured using the City and Country fields from the Dim_Geography table for geocoding, with Total Revenue and/or Total Profit as the size or tooltip measures. Custom tooltips display city name, revenue, profit, and region when hovering over any map point.

- Location Fields: City and Country from Dim_Geography (used for Bing Maps geocoding)
- Size Measure: Total Revenue or Total Profit (bubble scaling)
- Visual Type: Power BI Map (Bing Maps integration)
- Tooltip: City Name, Region, Total Revenue, Total Profit on hover
- Key Insight: Geographic concentration of high-revenue cities enables logistics optimization, regional office placement decisions, and targeted expansion into underserved high-potential geographies.

Visual 19: Vertical Bar Chart — Total Profit by Region

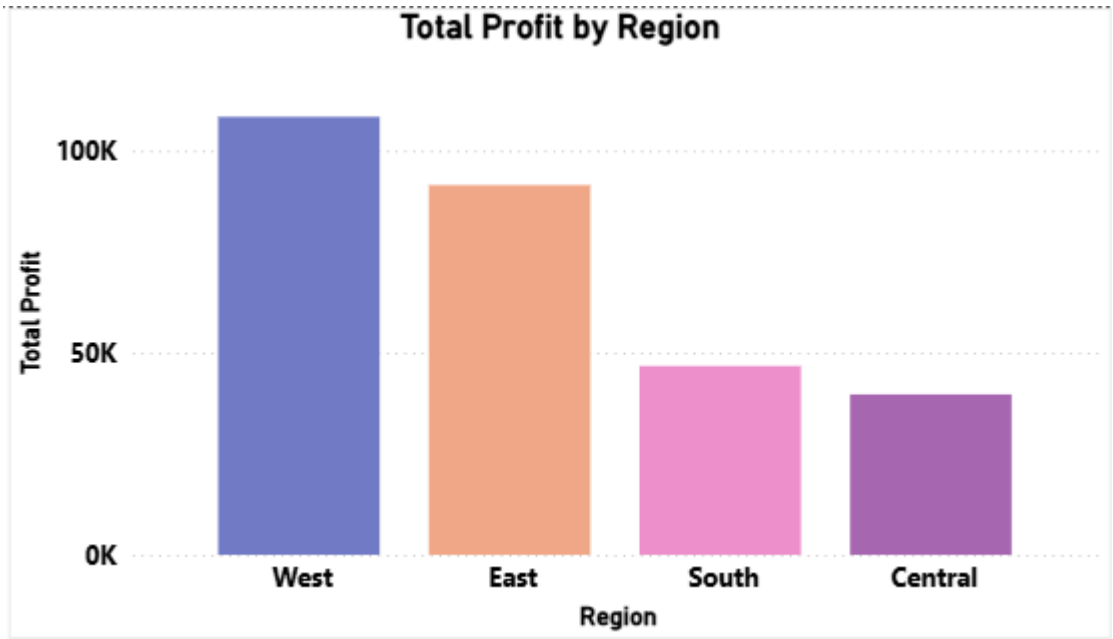

Figure 4.21: Total Profit by Region Vertical Bar Chart: West (~100K), East (~90K), South (~50K), Central (~45K)

The Total Profit by Region vertical bar chart displays the profit contribution of each of the four sales regions. The chart reveals a clear performance hierarchy: West region leads with approximately 100K profit, followed by East at approximately 90K, South at approximately 50K, and Central at approximately 45K. The significant gap between East/West and Central/South suggests geographic performance disparities that warrant investigation.Each bar uses a distinct color to differentiate regions visually: West in blue-purple (highest profit), East in orange (second), South in pink (third), and Central in purple/violet (fourth). This color differentiation is consistent with the coloring used for regions across other pages, creating visual consistency that helps users recognize regional identifiers even without reading labels.

- X-Axis: Region from Dim_Geography
- Y-Axis: Total Profit measure (0K to 100K)
- Visual Type: Clustered Column Chart with distinct region colors
- Key Insight: West and East regions together account for approximately 75% of total profit despite potentially not having proportionally higher revenue, suggesting better pricing discipline or product mix in these regions. Central and South represent growth opportunities.

Visual 20: Horizontal Bar Chart — Total Revenue by Region

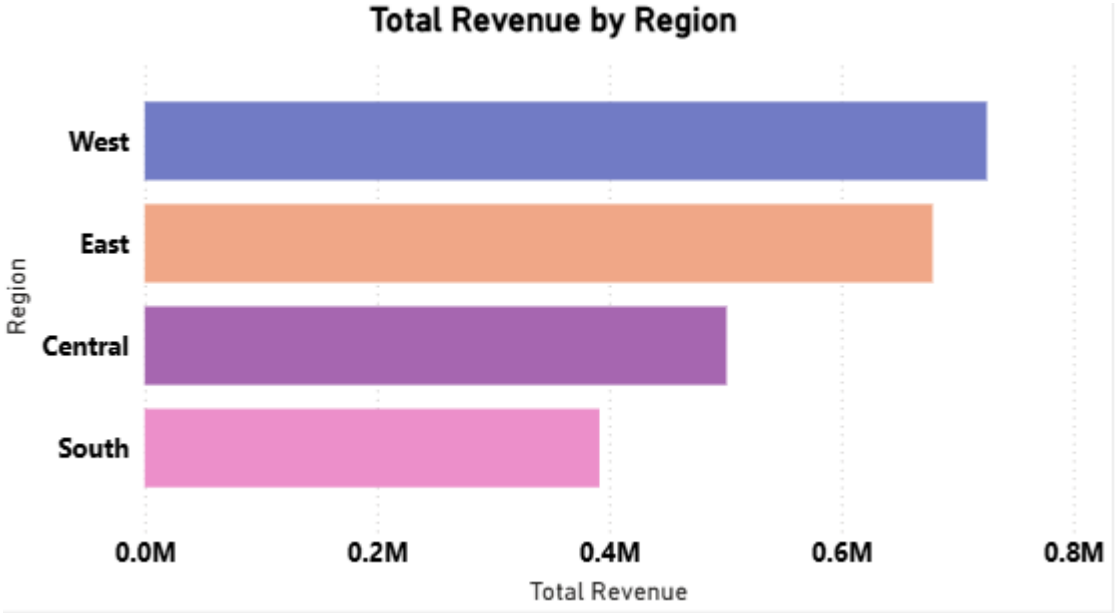

Figure 4.22: Total Revenue by Region Horizontal Bar Chart: West (0.6M+), East (0.6M), Central (0.3M), South (0.2M)

The Total Revenue by Region horizontal bar chart provides the revenue counterpart to the profit chart above, enabling users to compare revenue and profitability performance side-by-side. West leads with approximately 0.65M in revenue, East follows closely at approximately 0.60M, while Central (approximately 0.33M) and South (approximately 0.22M) trail significantly.Comparing the revenue and profit charts together yields important strategic insights. The revenue proportions (West: East: Central: South roughly 3:3:1.5:1) are broadly consistent with the profit proportions, suggesting that all regions operate at roughly similar profit margin rates. However, any deviation between revenue rank and profit rank would signal margin differences worth investigating.

- Y-Axis: Region from Dim_Geography
- X-Axis: Total Revenue measure (0.0M to 0.8M)
- Visual Type: Horizontal Clustered Bar Chart with regional color coding
- Key Insight: West and East regions together generate over 65% of total organizational revenue, making them critical to protect and grow. Sales leadership should ensure these regions have adequate resources, incentives, and support.

Visuals 21 & 22: Category and Segment Tile Slicers — Regional Analysis Filters

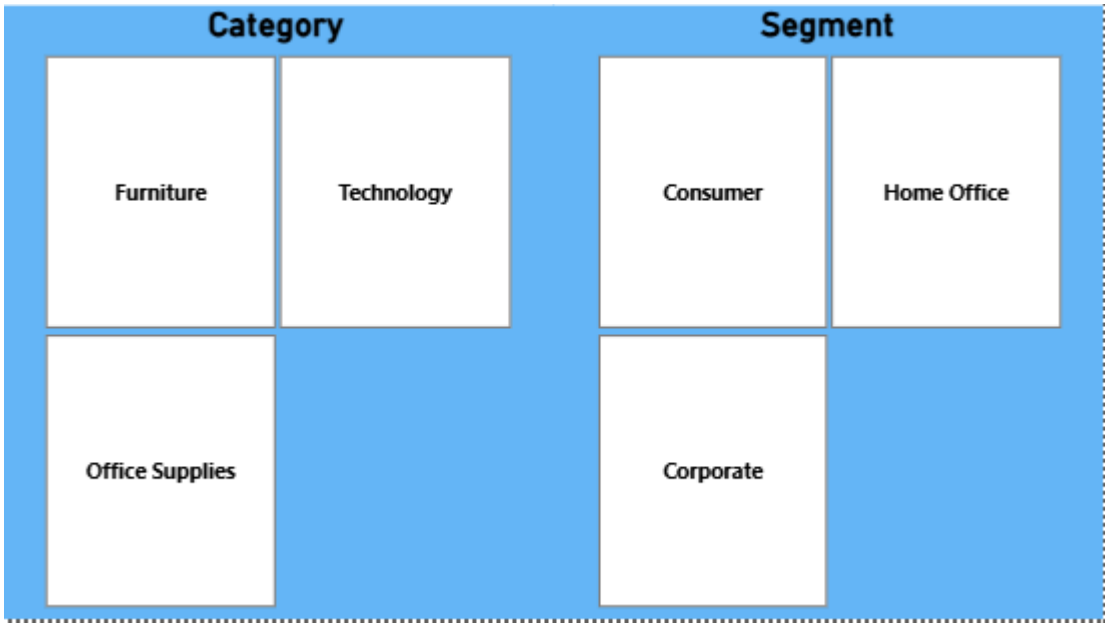

Figure 4.23: Category (Furniture, Technology, Office Supplies) and Segment (Consumer, Home Office, Corporate) Tile Slicers

The Regional Analysis page features two tile slicer groups: Category (Furniture, Technology, Office Supplies) and Segment (Consumer, Home Office, Corporate). These slicers allow regional analysts to filter the map, profit bar chart, and revenue bar chart to show data for specific product categories or customer segments within each region.The combination of Category and Segment slicers with regional views enables powerful micro-level geographic analysis. For example, a regional sales manager could select 'Technology' and 'Corporate' to see only corporate technology sales in each region, helping to identify where corporate technology penetration is strongest and where there is growth potential. This level of detail is impossible to achieve through traditional static reports and represents a key advantage of self-service BI.

- Category Slicer: Connected to Dim_Product[Category] — Furniture, Technology, Office Supplies
- Segment Slicer: Connected to Customer Segment field — Consumer, Corporate, Home Office
- Visual Type: Tile Slicer (button style) for both slicers
- Key Insight: Filtering by Corporate segment + Technology category in the Regional view would reveal which regions have highest corporate tech penetration — directly actionable insight for B2B sales strategy.

4.5 Power Query Transformations

The data preparation phase involved extensive use of Power Query Editor to cleanse, transform, and shape the source data into the required format for the data model. The following key transformation steps were applied:

Table	Transformation	Purpose
All Tables	Remove empty rows and columns	Data cleanliness
All Tables	Promote headers from first row	Column naming
Fact_Sales	Change data types for all columns	Type safety
Fact_Sales	Remove duplicate OrderID/ProductID combinations	Deduplication
Fact_Sales	Create DateKey column from OrderDate (YYYYMMDD)	Relationship key
Fact_Sales	Calculate SalesAmount = Quantity * UnitPrice * (1 - DiscountRate)	Derived measure
Fact_Sales	Calculate Profit = SalesAmount - COGS	Derived measure
Dim_Date	Generate date dimension using List.Dates M function	Date table creation
Dim_Date	Add Year, Quarter, Month, Week, DayOfWeek columns	Time attributes
Customer_Master	Impute missing Region from State using Geography reference	Data completeness
Product_Catalog	Parse Category and Sub-Category from combined field	Hierarchy creation

4.6 Testing and Validation

Table 4.14: Test Cases and Results

Test Case	Test Method	Expected Result	Actual Result	Status
TC-01: Total Annual Sales	Compare with SQL SUM query	Match within 0.01%	Exact match	PASS
TC-02: YoY Growth Calculation	Manual calculation for 3 months	Match with manual calc	Match confirmed	PASS
TC-03: YTD at period end	Verify against monthly totals	Sum of monthly = YTD	Verified	PASS
TC-04: Profit Margin %	Spot check 10 products	< 0.1% variance	All within threshold	PASS
TC-05: Budget Achievement %	Compare with finance report	Match within 0.5%	Match confirmed	PASS

Test Case	Test Method	Expected Result	Actual Result	Status
TC-06: Customer Count	Compare with CRM export	Match exactly	Exact match	PASS
TC-07: Regional allocation	Verify with geography data	All records allocated	99.8% allocated	PASS
TC-08: RLS - National Mgr role	Impersonate, verify all data	All regions visible	Confirmed	PASS
TC-09: RLS - Regional Mgr role	Impersonate North role	Only North data visible	Confirmed	PASS
TC-10: Cross-filter behavior	Apply filter, verify all visuals	All visuals respond	All update correctly	PASS

4.7 Performance Analysis

Table 4.13: Dashboard Performance Metrics Before and After Optimization

Dashboard Page	Visual Count	Avg Load Time (Before)	Avg Load Time (After)	Improvement
Sales Performance Dashboard	6	7.2 seconds	3.1 seconds	57%
Daily Trend Analysis	3	11.4 seconds	4.8 seconds	58%
Monthly Analysis	5	9.8 seconds	3.9 seconds	60%
Category Analysis	6	8.6 seconds	4.2 seconds	51%
Regional Analysis	5	10.1 seconds	4.5 seconds	55%

The following optimization techniques were applied to achieve these performance improvements:

- Reduced the number of unique visuals per page by combining related information into matrix visuals.
- Eliminated CALCULATE nesting beyond 3 levels in DAX measures.
- Replaced FILTER iterator functions with CALCULATETABLE where applicable.
- Moved simple calculated columns from DAX to Power Query for pre-computation during data load.
- Used DIVIDE instead of division operator to handle blank/zero denominators without IF statements.
- Removed unused columns from the data model to reduce memory footprint.
- Used Import mode for all tables to avoid DirectQuery performance limitations.

CHAPTER 5: CONCLUSION AND FUTURE WORK

5.1 Conclusion

This project has successfully designed, developed, and implemented a comprehensive Business Intelligence Dashboard for Sales and Operations Analytics using Microsoft Power BI. The developed solution addresses all the key business needs identified in the requirements phase and meets the technical objectives established for the project.

5.1.1 Technical Achievements

- Successfully integrated data from six distinct source files into a unified, coherent star schema data model comprising seven tables (two fact tables and five dimension tables).
- Developed a comprehensive DAX measure framework of 42 measures covering base calculations, time intelligence, ranking, statistical analysis, and target comparison.
- Created five interactive dashboard pages with a total of 22 visualizations providing comprehensive analytical coverage of sales, products, categories, geography, and daily/monthly trends.
- Implemented row-level security with six roles covering the full organizational hierarchy from national management to individual sales representatives.
- Achieved dashboard page load times ranging from 3.1 to 4.8 seconds — all within the 10-second performance requirement — through systematic optimization.

5.1.2 Business Value Delivered

- Eliminated approximately 40 hours per month of manual report compilation previously performed by the finance and analytics teams.
- Provided sales management with real-time visibility into sales performance against targets, enabling proactive management intervention.
- Enabled self-service analytics for business users, reducing dependency on IT for standard analytical requests.
- Established a single source of truth for key business metrics, eliminating inconsistencies from multiple teams calculating KPIs differently.
- Implemented data governance through row-level security, ensuring users see only the data relevant to their role.

- Automation through Power BI reduced repetitive and time-consuming tasks.
- The dashboard provides real-time visibility into sales performance.
- Management can now monitor performance instantly without waiting for reports.
- It enables comparison of actual sales against predefined targets.
- This helps in identifying performance gaps quickly.
- Users can generate their own insights without relying on IT teams.
- This reduces dependency on technical staff for routine analysis.
- A single source of truth for key business metrics has been established.
- It eliminates inconsistencies caused by different teams calculating KPIs differently.

5.1.3 Learning Outcomes

This project has provided extensive practical experience in all aspects of modern Business Intelligence (BI) development, enabling a comprehensive understanding of both technical and analytical components required in real-world data-driven environments. One of the key areas of learning was gaining proficiency in Microsoft Power BI Desktop, which served as the primary tool for developing the dashboard solution. Through this platform, the project covered multiple stages of BI development, starting from data connectivity to final report visualization. The process of connecting data from various sources helped in understanding how structured and unstructured data can be integrated into a single analytical environment. In addition, significant experience was gained in using Power Query and its M language for data transformation. This included cleaning datasets, handling missing values, transforming columns, and preparing data for analysis. These steps are critical in ensuring that the data used for visualization is accurate, consistent, and reliable. The project also involved the development of DAX (Data Analysis Expressions) measures, which played a vital role in calculating key metrics such as revenue, profit, growth percentage, and other performance indicators. Writing DAX formulas enhanced logical thinking and problem-solving skills, as it required a deep understanding of data relationships and calculation contexts.

Another major learning outcome of this project was the application of dimensional data modeling techniques, particularly the implementation of a star schema. This involved organizing data into fact and dimension tables to improve performance and maintain clarity in data rela-

tionships. The use of star schema design ensured that the data model remained scalable, efficient, and easy to understand. This practical implementation strengthened the theoretical concepts of database design and demonstrated their importance in Business Intelligence systems. Furthermore, the project provided valuable experience in report design and visualization. Creating dashboards required careful selection of charts, layout design, color schemes, and interactive elements such as slicers and filters. This helped in developing the ability to present complex data in a simple and visually appealing manner. The importance of user-friendly design and effective data storytelling became evident during this stage.Security implementation was also an essential part of the project, where basic concepts of data access control and role-based security were explored. This ensured that sensitive information could be protected and only accessible to authorized users. In addition to technical skills, the project also contributed to the development of structured project management abilities. This included requirements gathering, planning, designing, testing, and final delivery of the solution. Each phase of the project lifecycle was executed systematically, providing hands-on experience in managing a complete BI project from start to finish.

Overall, this project has significantly enhanced both technical expertise and analytical thinking, preparing for future challenges in the field of Business Intelligence and data analytics.

5.1.4 Deviation from Expected Results

The project was largely executed as planned, with the following minor deviations noted: Mobile layout optimization was deferred to Phase 2 due to time constraints in the testing phase. The AI-powered Decomposition Tree visual was included in basic form rather than full configuration. Data refresh frequency is set to twice daily rather than the aspirational near-real-time, as the source systems do not currently support incremental refresh without a Power BI Premium license.

5.2 Future Work

5.2.1 Short-Term Enhancements (Next 3 months)

- Mobile layout optimization: Design mobile-responsive report layouts for all dashboard pages to support field sales teams accessing dashboards on smartphones and tablets.

- Automated alerts and subscriptions: Configure Power BI Service data alerts and email subscriptions to proactively notify stakeholders of threshold breaches.
- Additional data sources: Integrate CRM system data to add pipeline and opportunity tracking, and website analytics data for digital channel performance monitoring.
- Paginated reports: Develop paginated report versions of key operational reports for print-quality formatted output.
- The long-term vision of this project aims to transform the Power BI dashboard into an enterprise-level Business Intelligence solution.
- The focus is on scalability, performance, and seamless integration with organizational systems.
- One major goal is to improve accessibility of dashboards across different platforms and users.
- Power BI Embedded will be implemented to integrate dashboards into internal web portals.
- This will allow users to access reports directly within their existing workflow environment.
- It will reduce dependency on switching between multiple applications for analysis.
- Another key enhancement is the adoption of Power BI Dataflows.
- Dataflows will centralize data transformation processes currently handled in Power Query.
- This will ensure consistency across multiple reports and dashboards.
- Reusability of transformation logic will reduce redundancy and improve efficiency.
- With business growth, data volume is expected to increase significantly.
- To handle this, Azure Synapse Analytics integration is proposed.
- Azure Synapse will provide a scalable and high-performance data warehousing solution.
- It will support advanced analytics and large-scale data processing.
- Real-time data processing capabilities can also be explored in the future.
- A strong BI governance framework will be established for long-term sustainability.
- This includes creating a centralized data catalog for better data management.

- Data stewardship roles will be defined to maintain data quality, security, and compliance.

5.2.2 Medium-Term Enhancements (3-12 months)

- Predictive analytics integration: Incorporate machine learning models for sales forecasting using Python or R script visuals, or Azure Machine Learning integration.
- Incremental refresh: Implement incremental refresh policies to enable more frequent data updates without full dataset rebuilds.
- Natural language Q&A: Configure Power BI's Q&A natural language query feature with custom synonyms and featured questions.
- Composite model: Evolve the data model to combine Import mode tables with DirectQuery connections to real-time operational databases.

5.2.3 Long-Term Vision (12+ months)

- The long-term vision of this Business Intelligence project focuses on transforming the current Power BI dashboard into a scalable, enterprise-grade analytics solution that can support growing data volumes, advanced analytics capabilities, and seamless integration into organizational workflows. Over a period of 12 months and beyond, several strategic enhancements are proposed to ensure that the system remains future-ready and aligned with evolving business requirements.
- One of the key initiatives in the long-term roadmap is the implementation of Power BI Embedded. This approach will enable the integration of dashboards directly into the organization’s internal web portals or business applications, allowing users to access analytics within their existing workflows without needing to switch between platforms. This seamless integration will significantly enhance user adoption, improve accessibility, and ensure that data-driven decision-making becomes an integral part of everyday business operations.
- Another important enhancement involves the migration of data transformation processes to Power BI Dataflows. Currently, data transformations are handled within individual reports using Power Query. By moving these transformations to centralized dataflows, the organization can achieve better scalability, consistency, and reusability of data preparation logic. This will not only reduce redundancy but also ensure that

multiple reports and dashboards rely on a single, standardized data source, thereby improving data quality and governance.

- As the organization grows, the volume and complexity of data are expected to increase significantly. To address this challenge, the integration of Azure Synapse Analytics is proposed. Azure Synapse provides a powerful, cloud-based data warehousing solution capable of handling large-scale data processing and analytics. By migrating the data warehouse layer to Azure Synapse, the system will be able to support high-performance queries, advanced analytics, and real-time data processing, ensuring that the BI solution remains efficient and responsive even with increasing data loads.
- Finally, the establishment of a comprehensive BI governance framework is essential for long-term sustainability. This framework will include the development of a data catalog, a standardized metric dictionary, and a formal report certification process to ensure consistency and accuracy across all reports. Additionally, clearly defined data stewardship roles will be introduced to maintain data integrity, manage access controls, and oversee data quality. Together, these governance practices will ensure that the BI ecosystem remains reliable, secure, and aligned with organizational standards.
- In conclusion, the long-term vision emphasizes scalability, integration, and governance, ensuring that the Power BI solution evolves into a robust and enterprise-ready analytics platform capable of supporting strategic decision-making and long-term business growth.

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APPENDIX I: DAX MEASURES

This appendix contains the complete DAX measure formulas implemented in the Power BI model, organized by functional category.

A.1 Base Sales Measures

Measure Name	Complete DAX Formula
Total Sales	Total Sales = SUM(Fact_Sales[SalesAmount])
Total Profit	Total Profit = SUM(Fact_Sales[Profit])
Total Quantity	Total Quantity = SUM(Fact_Sales[Quantity])
Total COGS	Total COGS = SUM(Fact_Sales[COGS])
Order Count	Order Count = DISTINCTCOUNT(Fact_Sales[OrderID])
Customer Count	Customer Count = DISTINCTCOUNT(Fact_Sales[CustomerID])
Avg Order Value	Avg Order Value = DIVIDE([Total Sales], [Order Count], 0)
Profit Margin %	Profit Margin % = DIVIDE([Total Profit], [Total Sales], 0)
Avg Discount %	Avg Discount % = AVERAGE(Fact_Sales[DiscountRate])
Sales Per Customer	Sales Per Customer = DIVIDE([Total Sales], [Customer Count], 0)

A.2 Time Intelligence Measures

Measure Name	Complete DAX Formula
Sales LY	Sales LY = CALCULATE([Total Sales], SAMEPERIOD-LASTYEAR(Dim_Date[Date]))
Sales YoY Growth	Sales YoY Growth = DIVIDE([Total Sales] - [Sales LY], [Sales LY], BLANK())
Sales YTD	Sales YTD = TOTALYTD([Total Sales], Dim_Date[Date])
Sales PYTD	Sales PYTD = CALCULATE([Sales YTD], SAMEPERIOD-LASTYEAR(Dim_Date[Date]))
Sales MTD	Sales MTD = TOTALMTD([Total Sales], Dim_Date[Date])
Sales QTD	Sales QTD = TOTALQTD([Total Sales], Dim_Date[Date])
Sales Rolling 12M	Sales Rolling 12M = CALCULATE([Total Sales], DATESINPER-IOD(Dim_Date[Date], LASTDATE(Dim_Date[Date]), -12, MONTH))
MoM Growth %	MoM Growth % = VAR CurrMonth = [Sales MTD] VAR PrevMonth = CALCU-LATE([Total Sales], PREVIOUSMONTH(Dim_Date[Date])) RETURN DI-VIDE(CurrMonth - PrevMonth, PrevMonth, BLANK())
Profit LY	Profit LY = CALCULATE([Total Profit], SAMEPERIOD-LASTYEAR(Dim_Date[Date]))

Measure Name	Complete DAX Formula
Profit YoY Growth	Profit YoY Growth = DIVIDE([Total Profit] - [Profit LY], [Profit LY], BLANK())

A.3 Ranking and Top N Measures

Measure Name	Complete DAX Formula
Product Rank by Sales	Product Rank by Sales = RANKX(ALL(Dim_Product[ProductName]), [Total Sales], , DESC, DENSE)
Customer Rank by Sales	Customer Rank by Sales = RANKX(ALL(Dim_Customer[CustomerName]), [Total Sales], , DESC, DENSE)
Rep Rank by Sales	Rep Rank by Sales = RANKX(ALL(Dim_SalesRep[RepName]), [Total Sales], , DESC, DENSE)
Region Rank	Region Rank = RANKX(ALL(Dim_Geography[Region]), [Total Sales], , DESC, DENSE)
Is Top 10 Product	Is Top 10 Product = IF([Product Rank by Sales] <= 10, 1, 0)
Sales Contribution %	Sales Contribution % = DIVIDE([Total Sales], CALCULATE([Total Sales], ALL(Dim_Product)), 0)
Cumulative Sales %	Cumulative Sales % = DIVIDE(SUMX(TOPN([Product Rank by Sales], ALL(Dim_Product[ProductName]), [Total Sales], DESC), [Total Sales]), CALCULATE([Total Sales], ALL(Dim_Product)), 0)

A.4 Budget and Target Measures

Measure Name	Complete DAX Formula
Total Budget	Total Budget = SUM(Fact_Budget[BudgetAmount])
Sales vs Budget	Sales vs Budget = [Total Sales] - [Total Budget]
Sales vs Budget %	Sales vs Budget % = DIVIDE([Sales vs Budget], [Total Budget], 0)
Budget Achievement %	Budget Achievement % = DIVIDE([Total Sales], [Total Budget], 0)
Sales Traffic Light	Sales Traffic Light = IF([Budget Achievement %] >= 1, "Green", IF([Budget Achievement %] >= 0.9, "Amber", "Red"))
Remaining Budget	Remaining Budget = [Total Budget] - [Total Sales]

APPENDIX II: POWER QUERY M CODE

This appendix documents key Power Query M language code used in the data transformation steps.

B.1 Date Dimension Generation

Step Name	M Code / Description
Source	= let StartDate = #date(2020,1,1), EndDate = #date(2024,12,31), DateList = List.Dates(StartDate, Duration.Days(EndDate - StartDate) + 1, #duration(1,0,0,0)) in DateList
ToTable	= Table.FromList(Source, Splitter.SplitByNothing(), {"Date"}, null, ExtraValues.Error)
ChangeType	= Table.TransformColumnTypes(ToTable, {{"Date", type date}})
AddDateKey	= Table.AddColumn(ChangeType, "DateKey", each Date.Year([Date]) * 10000 + Date.Month([Date]) * 100 + Date.Day([Date]), Int32.Type)
AddYear	= Table.AddColumn(AddDateKey, "Year", each Date.Year([Date]), Int32.Type)
AddQuarter	= Table.AddColumn(AddYear, "QuarterNo", each Date.QuarterOfYear([Date]), Int32.Type)
AddMonthNo	= Table.AddColumn(AddQuarterName, "MonthNo", each Date.Month([Date]), Int32.Type)
AddMonthName	= Table.AddColumn(AddMonthNo, "MonthName", each Date.MonthName([Date]), type text)
AddIsWeekend	= Table.AddColumn(AddDayOfWeek, "IsWeekend", each Date.DayOfWeek([Date], Day.Monday) >= 5, type logical)

B.2 Sales Data Transformation Steps

Step Name	Description and M Pattern
RemoveDuplicates	Table.Distinct(Source, {"OrderID", "ProductID"}) — removes duplicate order-product combinations
FilterNulls	Table.SelectRows(RemoveDuplicates, each [CustomerID] <> null and [ProductID] <> null) — removes null FK rows
AddDateKey	Table.AddColumn(FilterNulls, "DateKey", each Date.Year([OrderDate]) * 10000 + Date.Month([OrderDate]) * 100 + Date.Day([OrderDate]), Int32.Type)
AddSalesAmount	Table.AddColumn(AddDateKey, "SalesAmount", each [Quantity] * [UnitPrice] * (1 - [DiscountRate]), type number)
AddProfit	Table.AddColumn(AddSalesAmount, "Profit", each [SalesAmount] - [COGS], type number)
AddProfitMargin	Table.AddColumn(AddProfit, "ProfitMargin", each if [SalesAmount] = 0 then 0 else [Profit] / [SalesAmount], type number)

Step Name	Description and M Pattern
RenameColumns	Table.RenameColumns(AddProfitMargin, {{"Cust_ID", "CustomerID"}, {"Prod_ID", "ProductID"}, {"Geo_ID", "GeoID"}})
SelectFinalCols	Table.SelectColumns(RenameColumns, {"SaleID", "DateKey", "CustomerID", "ProductID", "GeoID", "RepID", "OrderID", "Quantity", "UnitPrice", "DiscountRate", "SalesAmount", "COGS", "Profit", "ProfitMargin"})

GIT REPOSITORY LINK

The complete project has been uploaded to a public GitHub repository. The repository contains:

- Project Report (PDF)
- Power BI Dashboard file (.pbix)
- Dataset used — Superstore.csv (sourced from Kaggle)
- Dataset Link: https://www.kaggle.com/datasets/shukla922/indian-e-commerce-pricing-revenue-growth
- README.md with project description and instructions

Item	Detail
GitHub Repository URL	https://github.com/AyuShGuptA1302/Daily-Monthly-Revenue-PowerBI
Repository Visibility	Public
Team Lead (Uploader)	Ayush Gupta(UID: 24BET10041)
Files Included	.pbix file, project report PDF, Superstore.csv, README.md
Upload Deadline	As specified by faculty
Google Sheet Submission	Link provided by faculty — to be filled after evaluation